

BFS Traversal

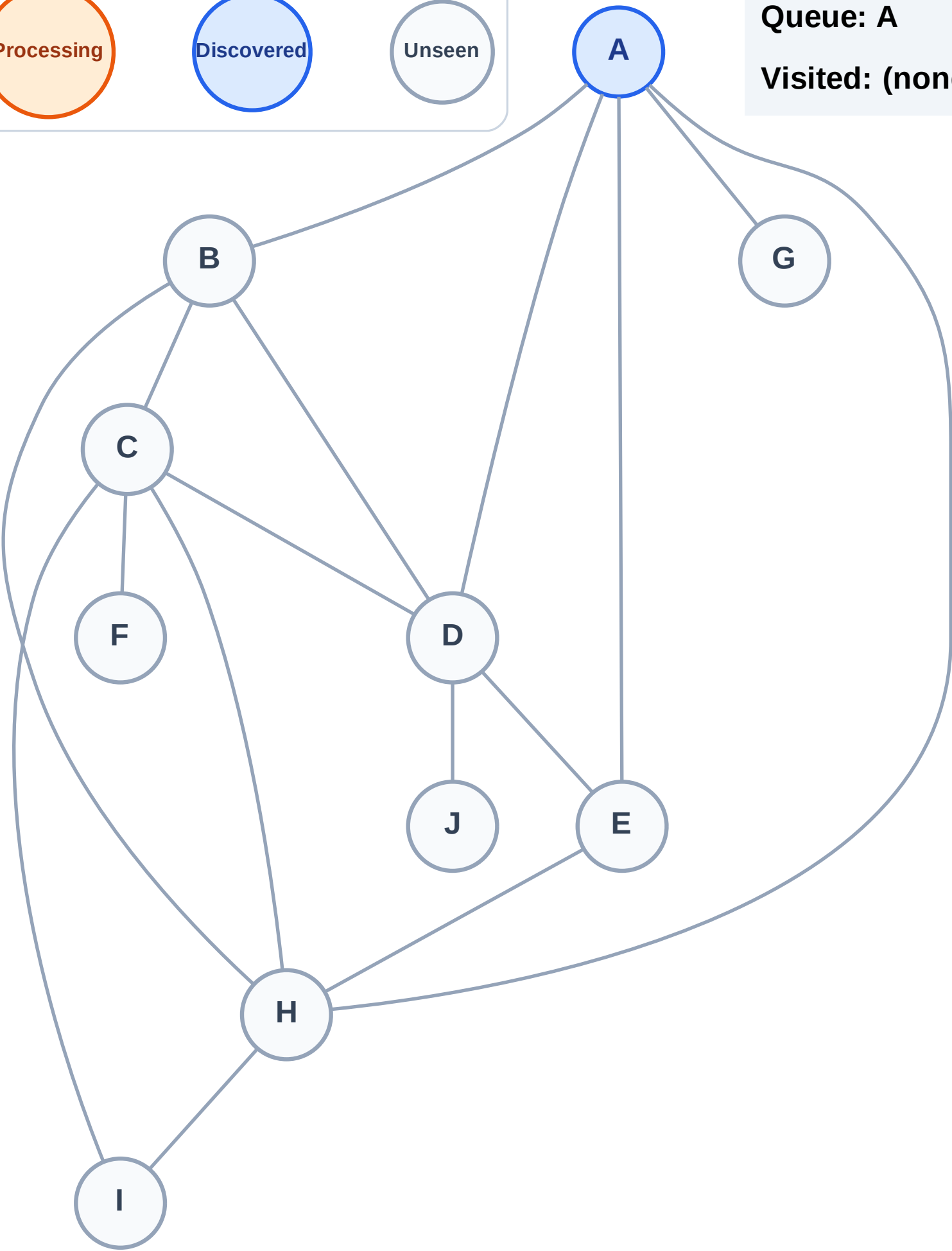
Step 1: Start at A

Legend



Queue: A

Visited: (none)



BFS Traversal

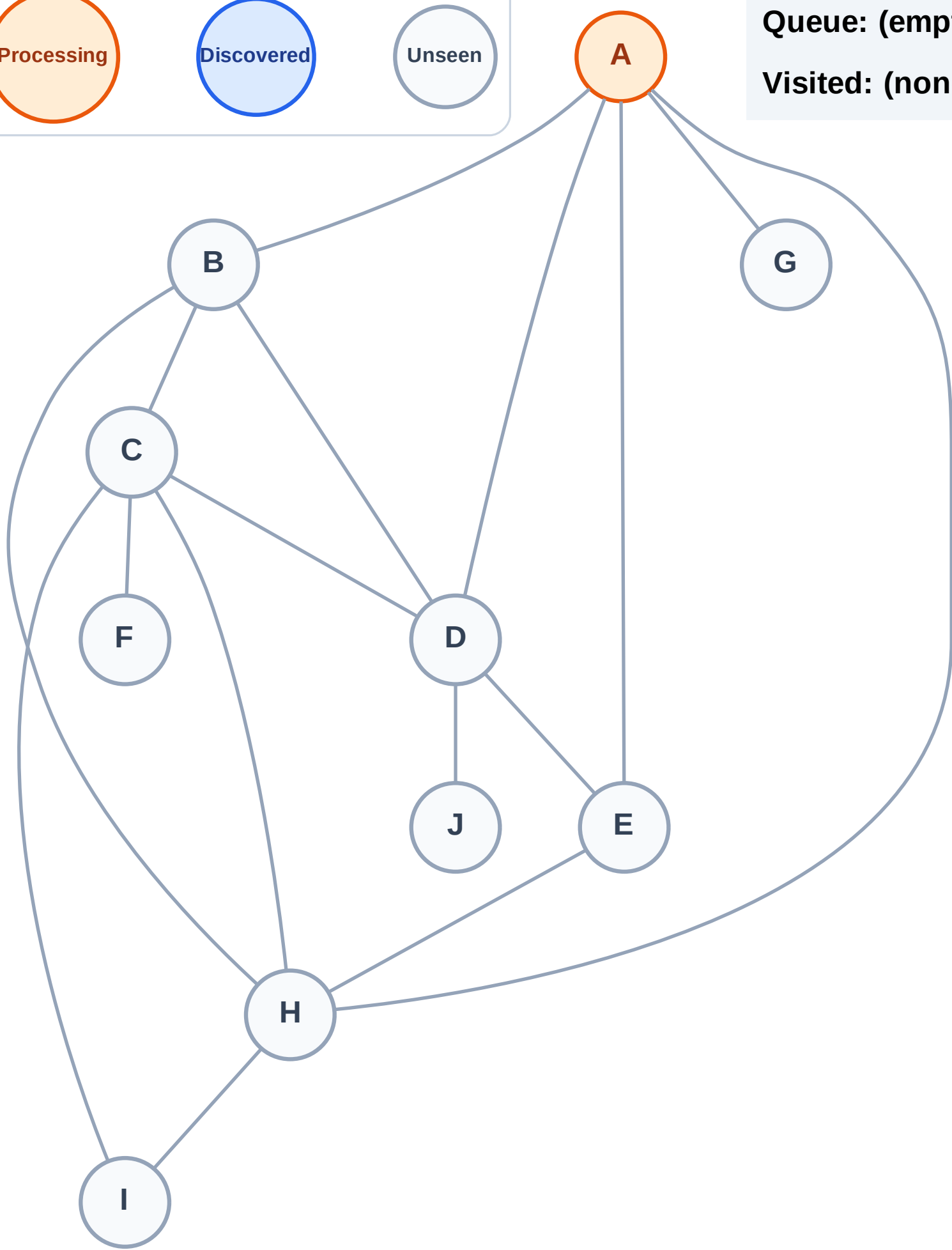
Step 2: Process node A

Legend



Queue: (empty)

Visited: (none)



BFS Traversal

Step 3: Discover B, D, E, G, H from A

Legend

Complete

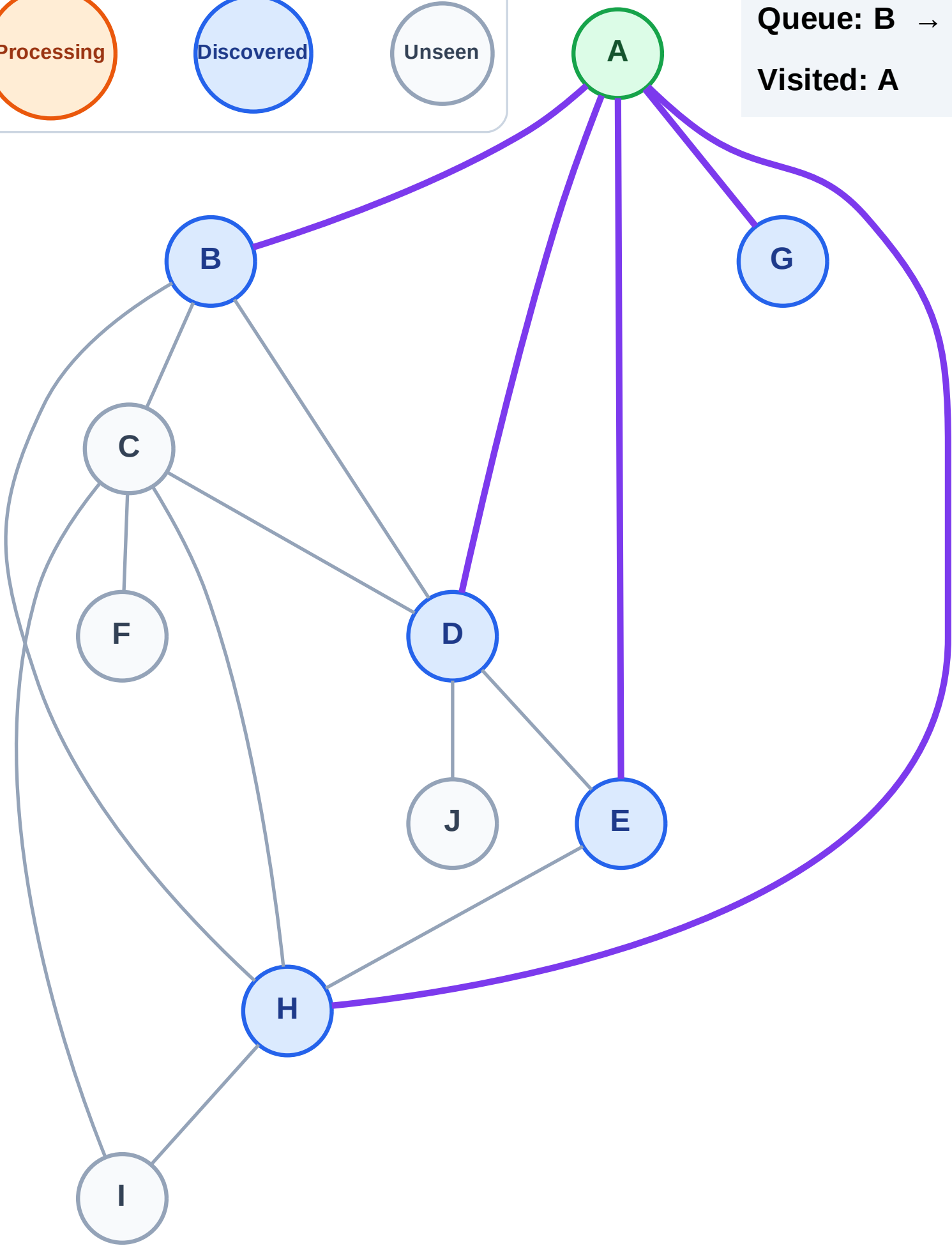
Processing

Discovered

Unseen

Queue: B → D → E → G → H

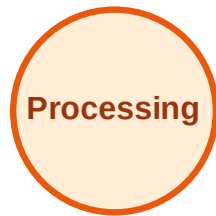
Visited: A



BFS Traversal

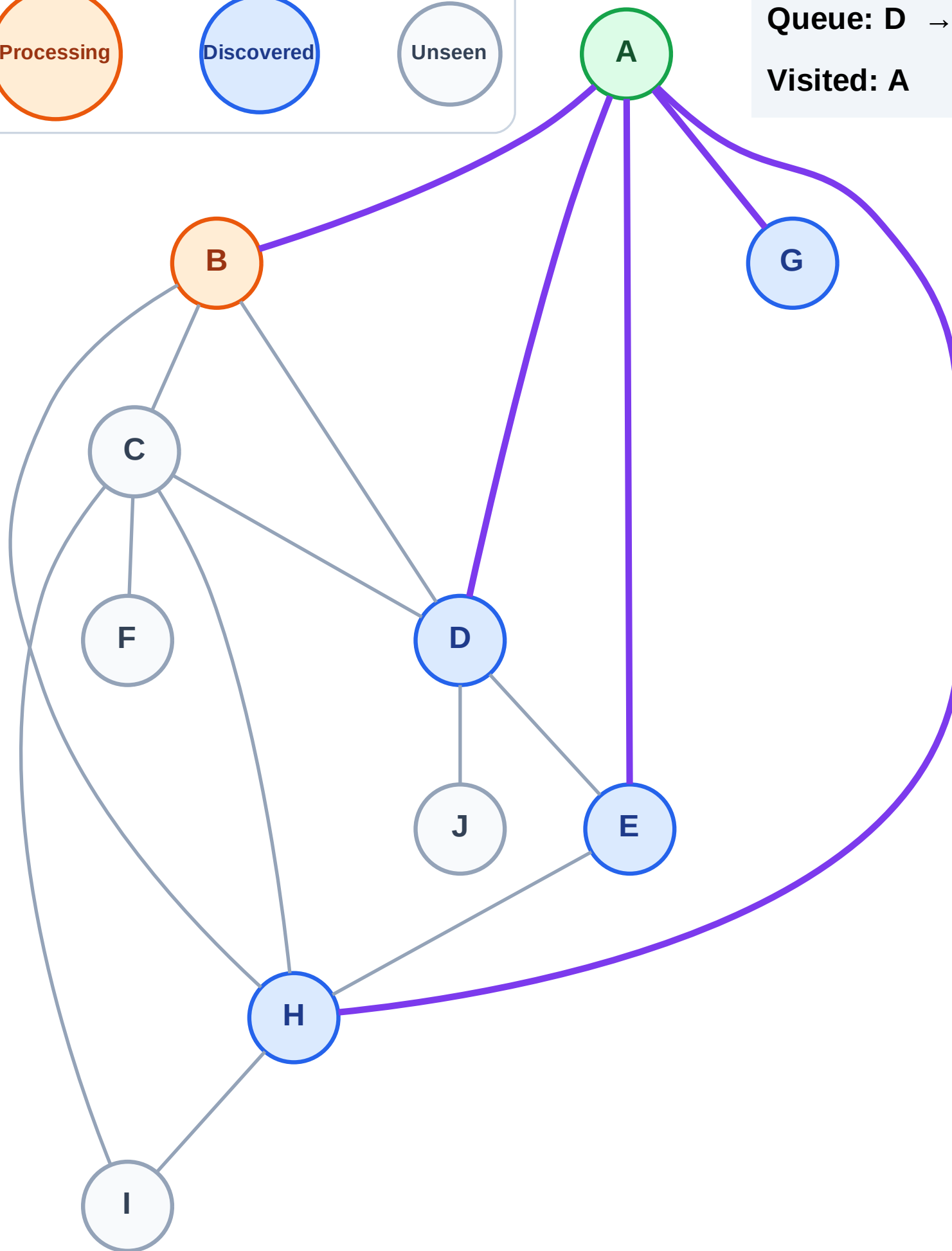
Step 4: Process node B

Legend



Queue: D → E → G → H

Visited: A



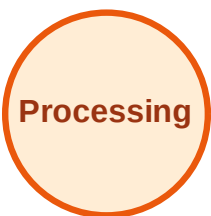
BFS Traversal

Step 5: Discover C from B

Legend



Complete



Processing



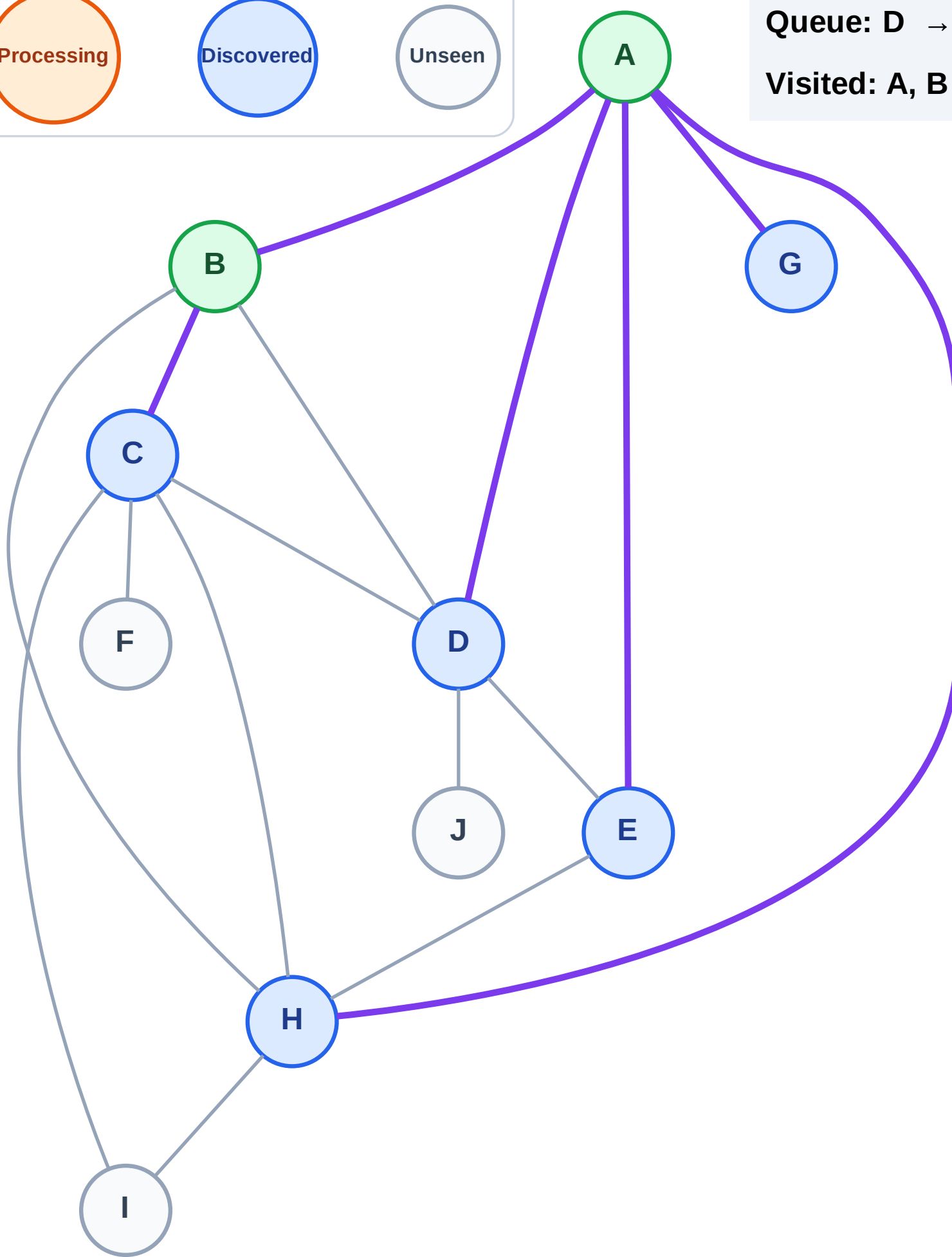
Discovered



Unseen

Queue: D → E → G → H → C

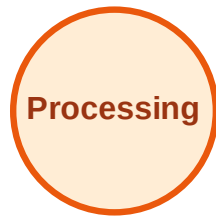
Visited: A, B



BFS Traversal

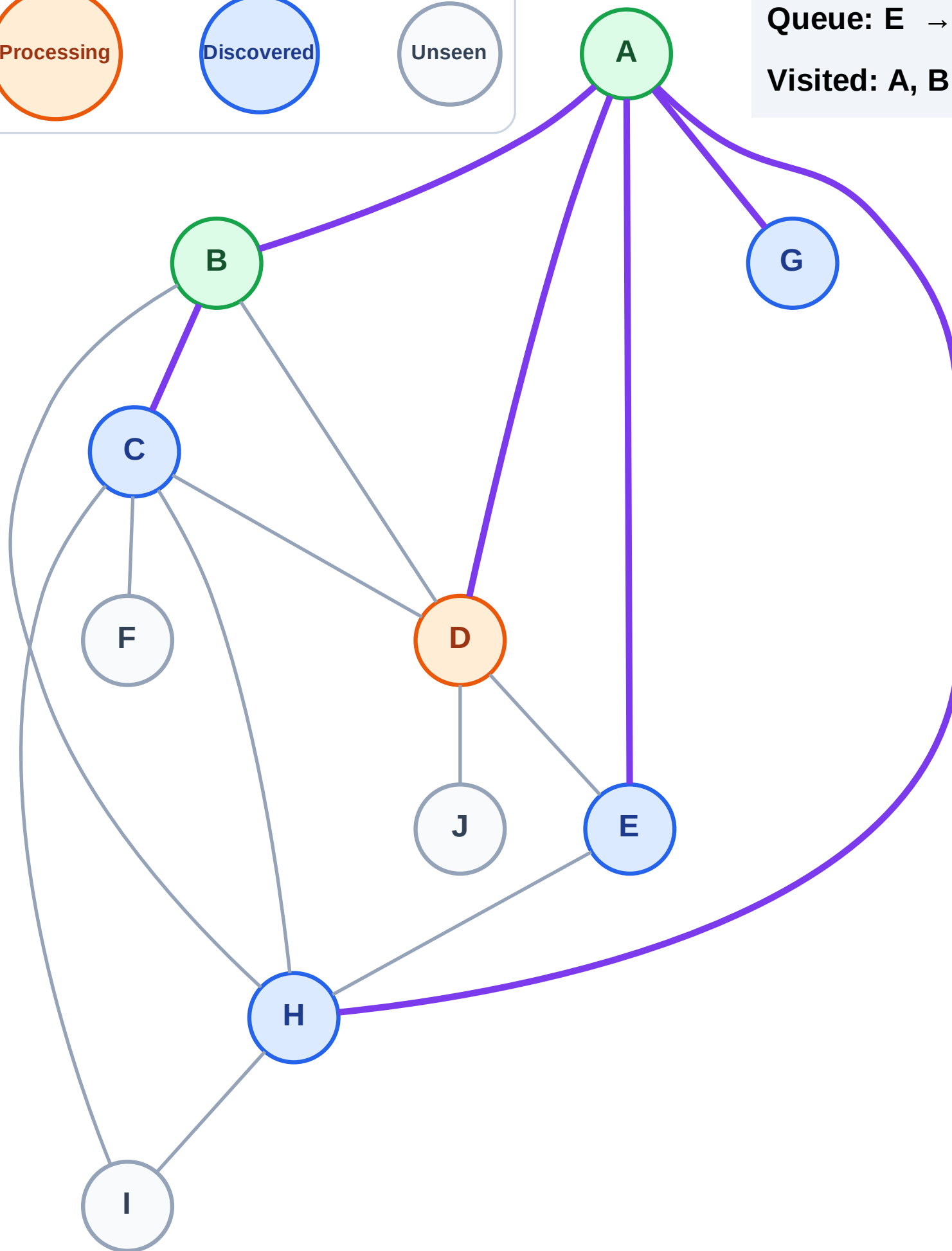
Step 6: Process node D

Legend



Queue: E → G → H → C

Visited: A, B



BFS Traversal

Step 7: Discover J from D

Legend



Complete



Processing



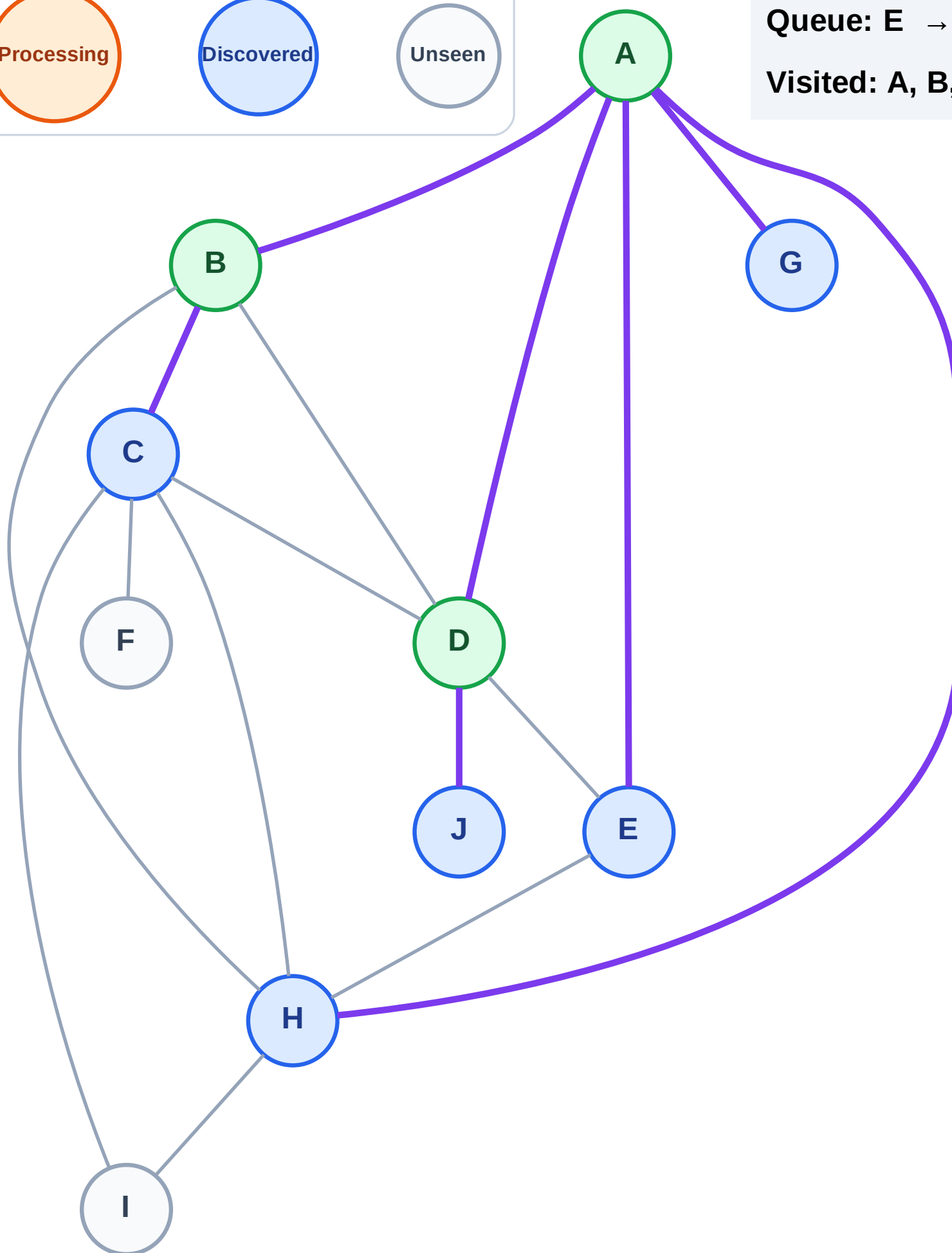
Discovered



Unseen

Queue: E → G → H → C → J

Visited: A, B, D



BFS Traversal

Step 8: Process node E

Legend

Complete

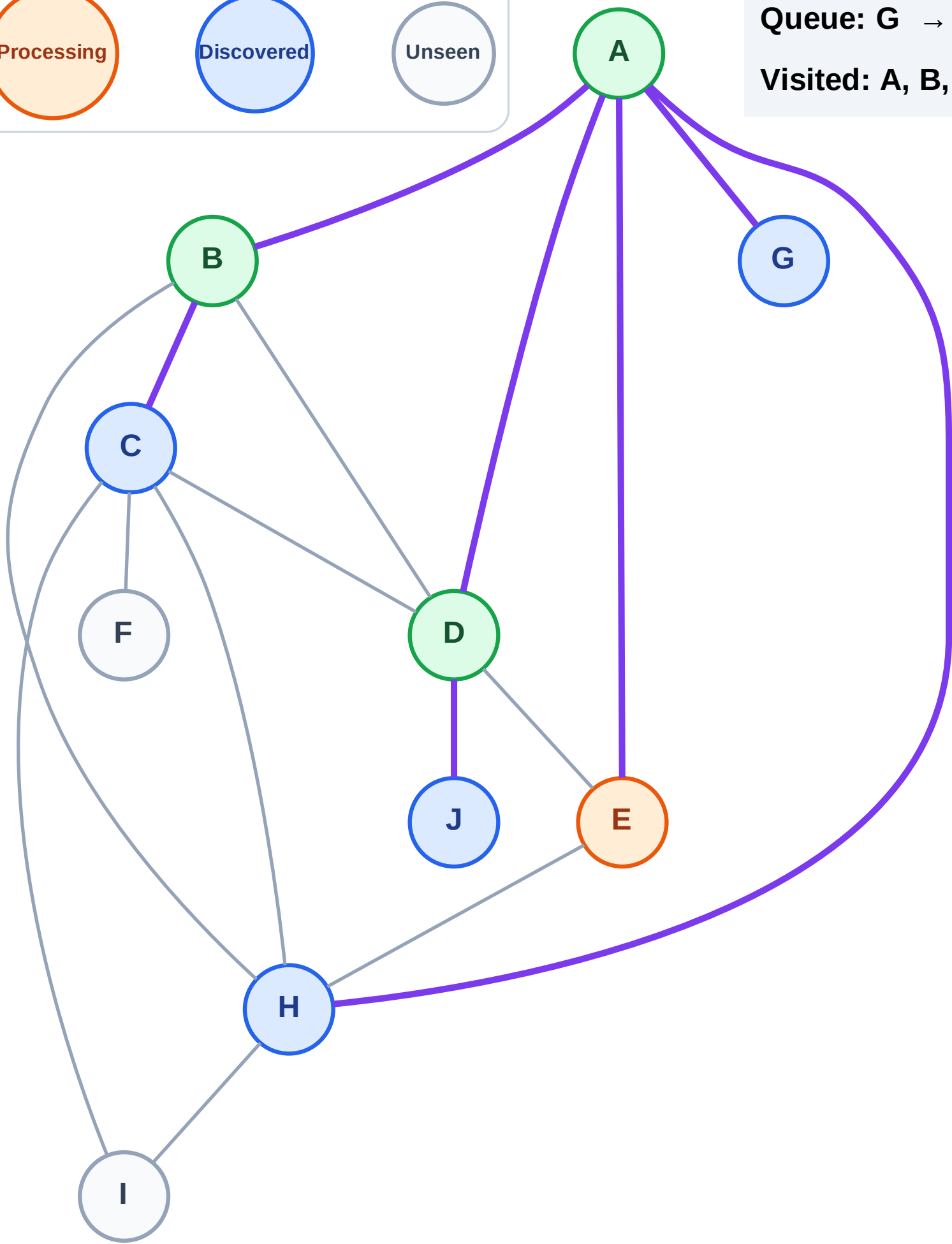
Processing

Discovered

Unseen

Queue: G → H → C → J

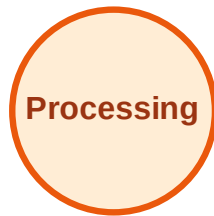
Visited: A, B, D



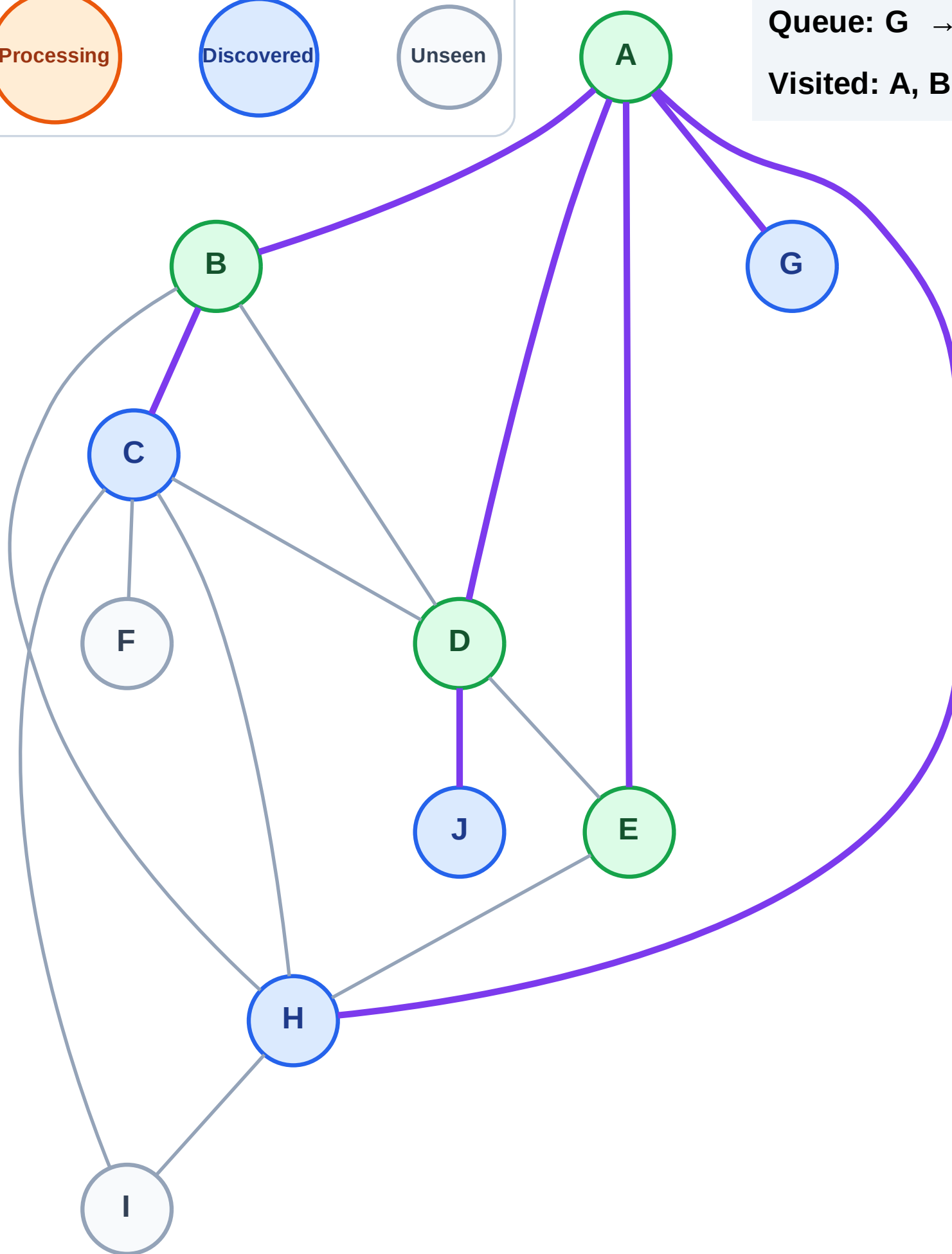
BFS Traversal

Step 9: No new neighbors from E

Legend



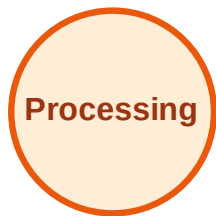
Queue: G → H → C → J
Visited: A, B, D, E



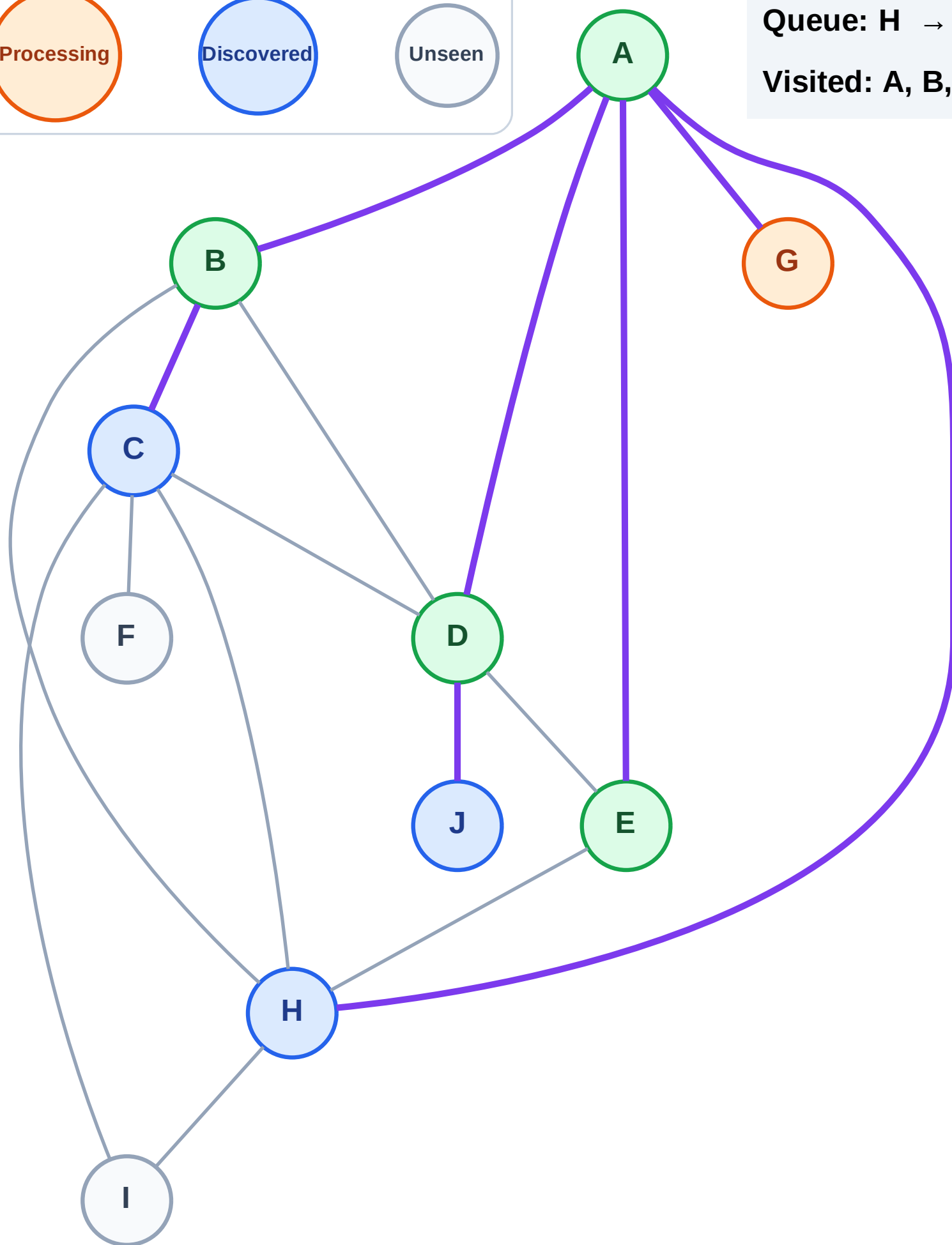
BFS Traversal

Step 10: Process node G

Legend



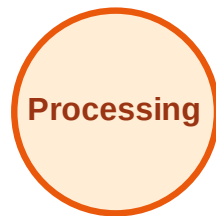
Queue: H → C → J
Visited: A, B, D, E



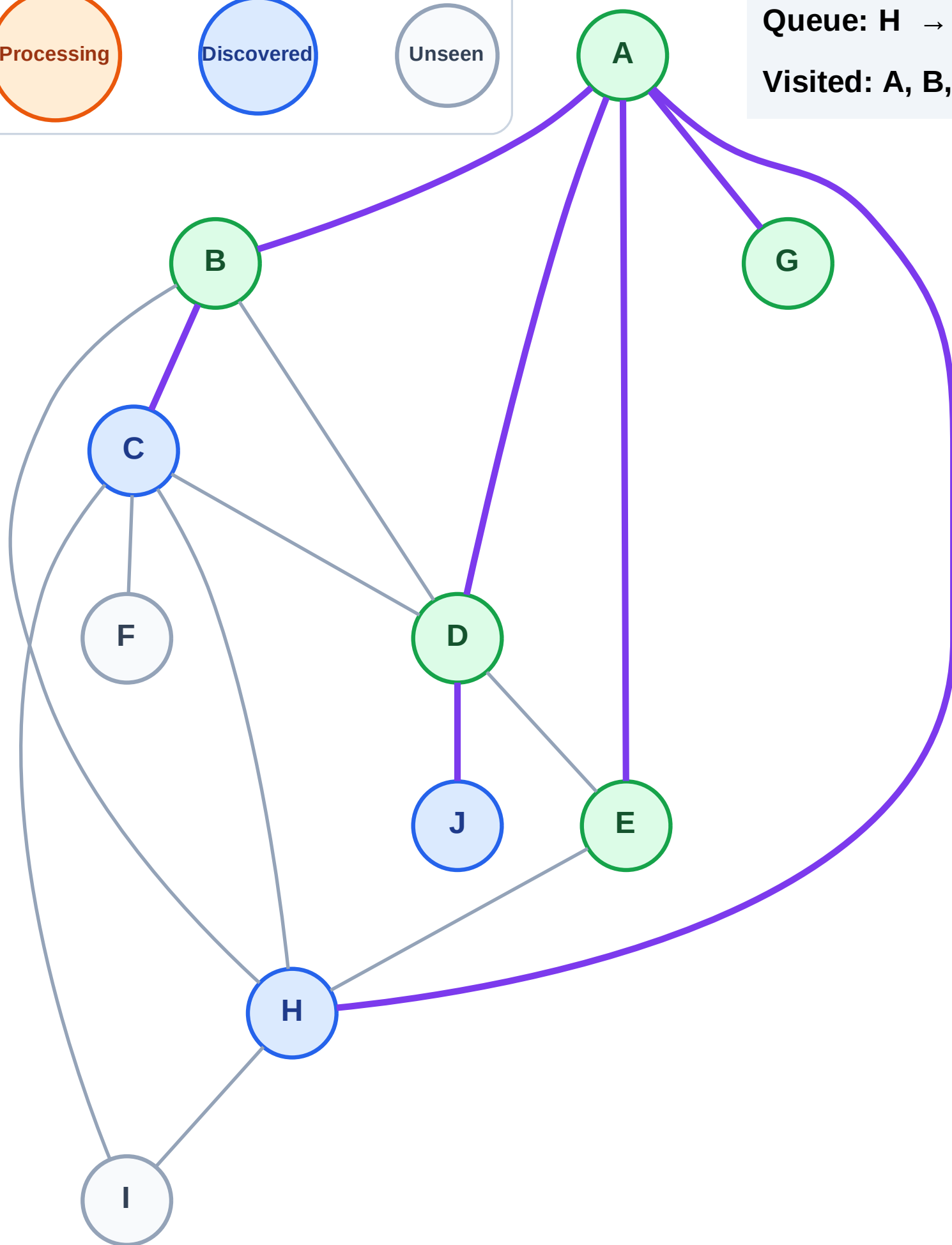
BFS Traversal

Step 11: No new neighbors from G

Legend

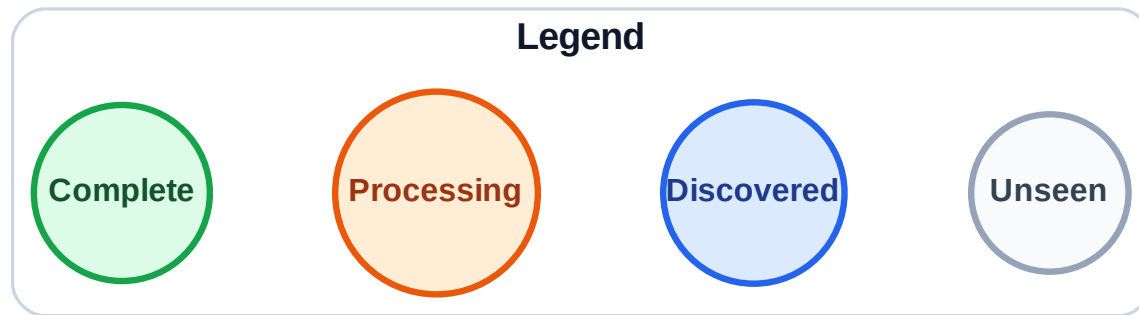


Queue: H → C → J
Visited: A, B, D, E, G

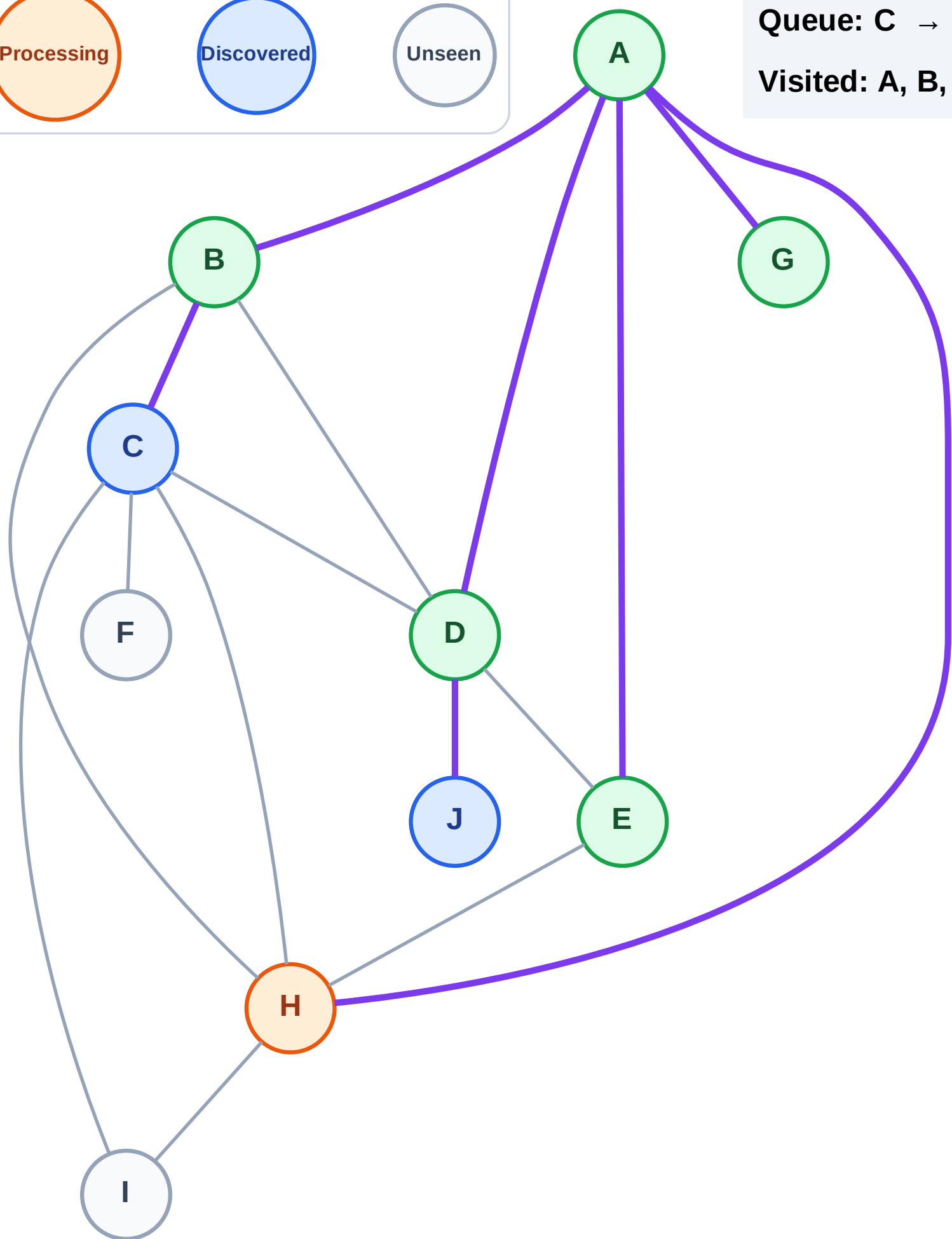


BFS Traversal

Step 12: Process node H

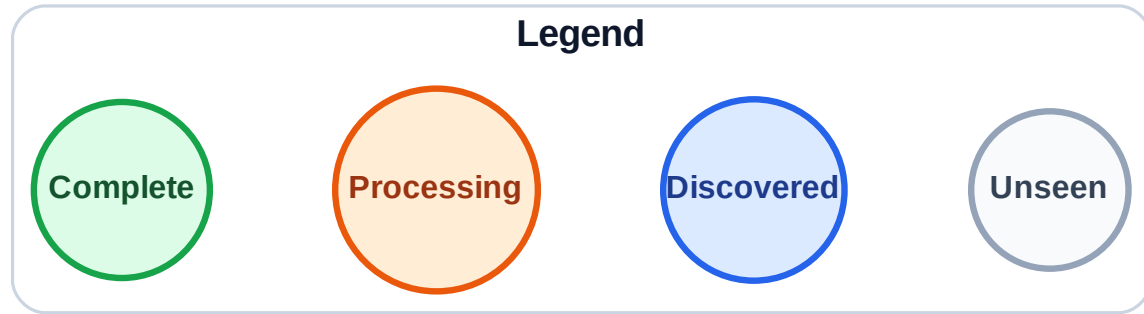


Queue: C → J
Visited: A, B, D, E, G

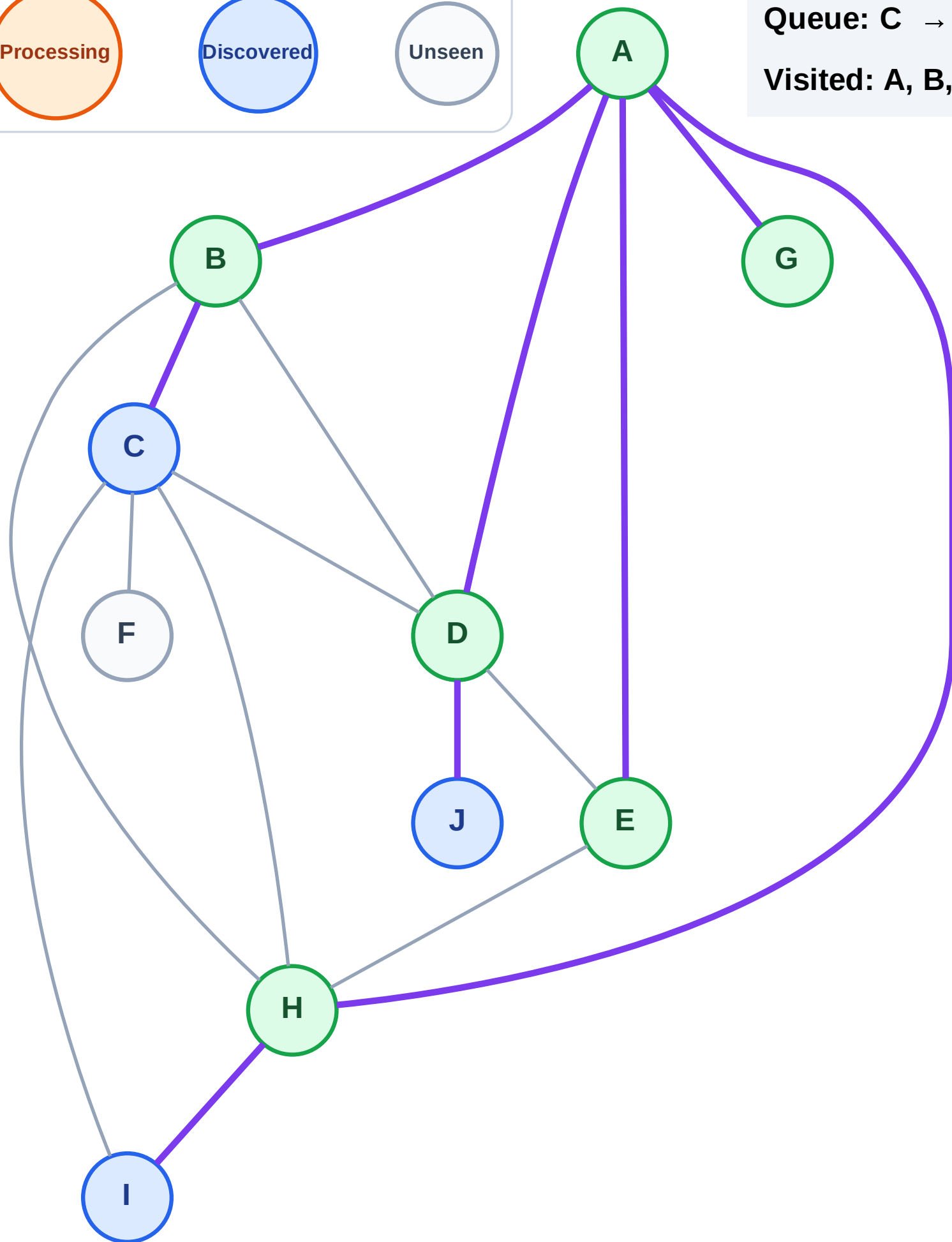


BFS Traversal

Step 13: Discover I from H



Queue: C → J → I
Visited: A, B, D, E, G, H



BFS Traversal

Step 14: Process node C

Legend

Complete

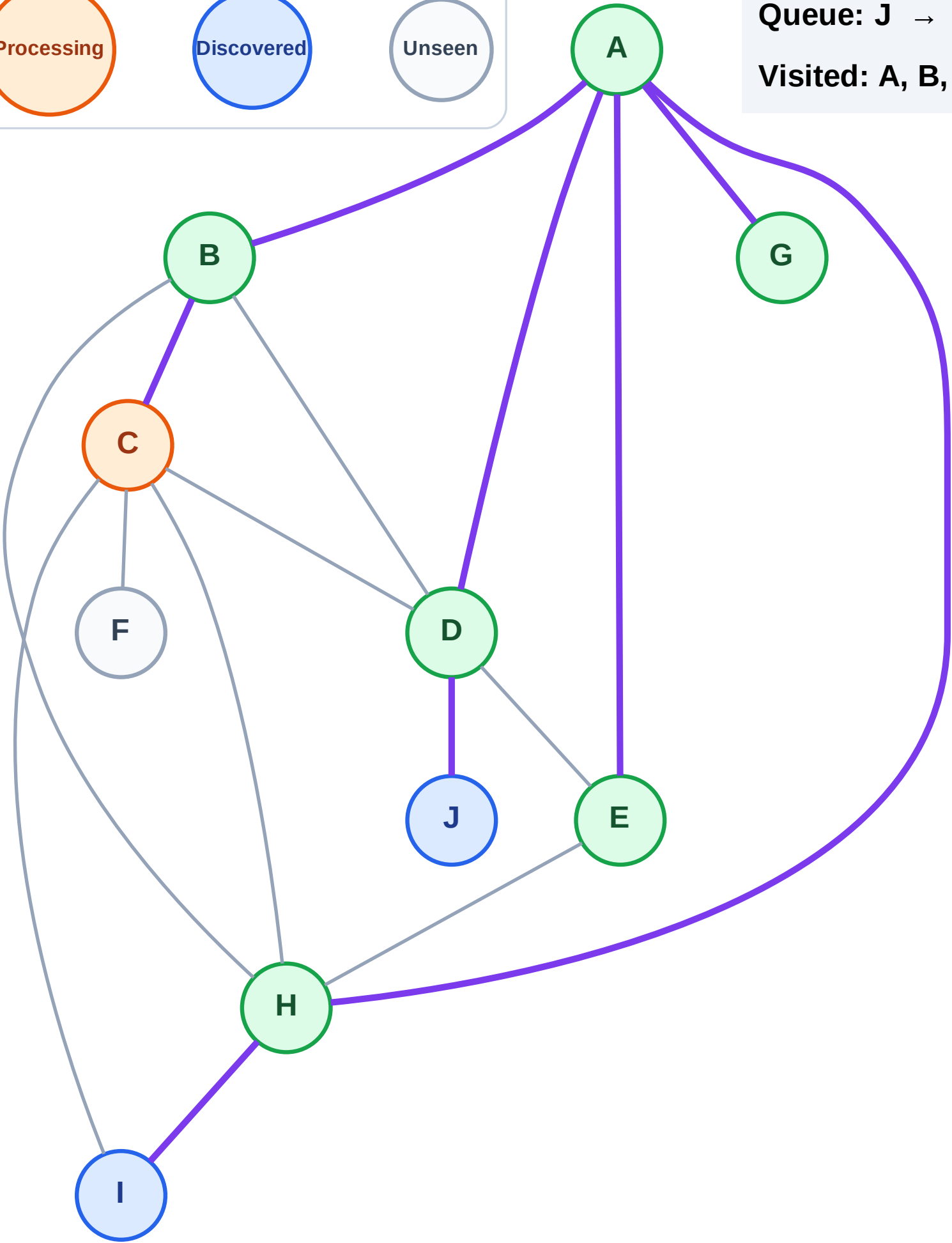
Processing

Discovered

Unseen

Queue: J → I

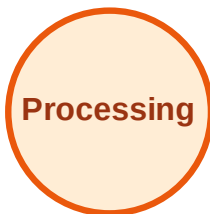
Visited: A, B, D, E, G, H



BFS Traversal

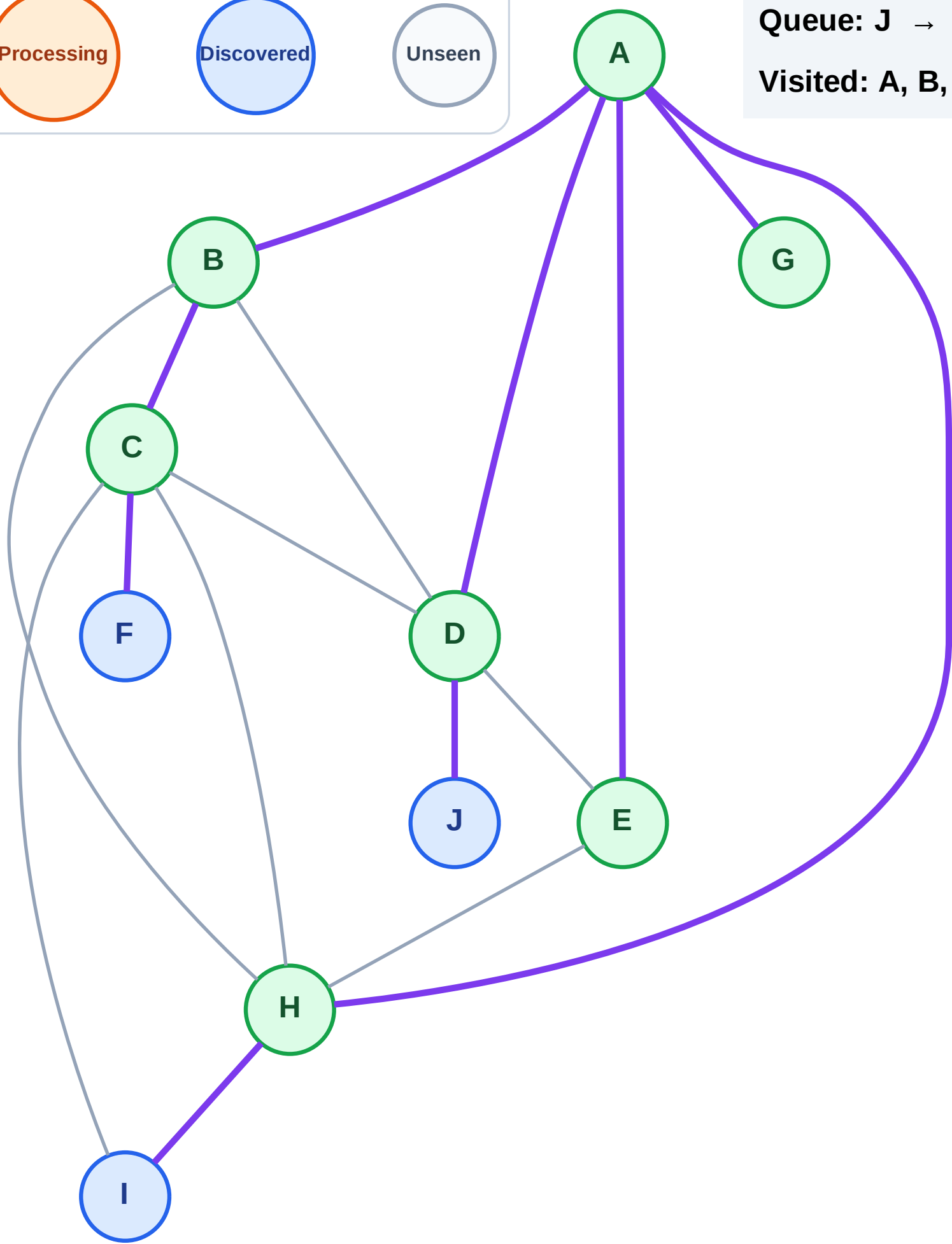
Step 15: Discover F from C

Legend



Queue: J → I → F

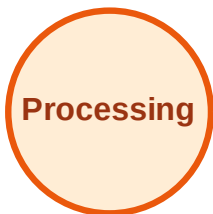
Visited: A, B, C, D, E, G, H



BFS Traversal

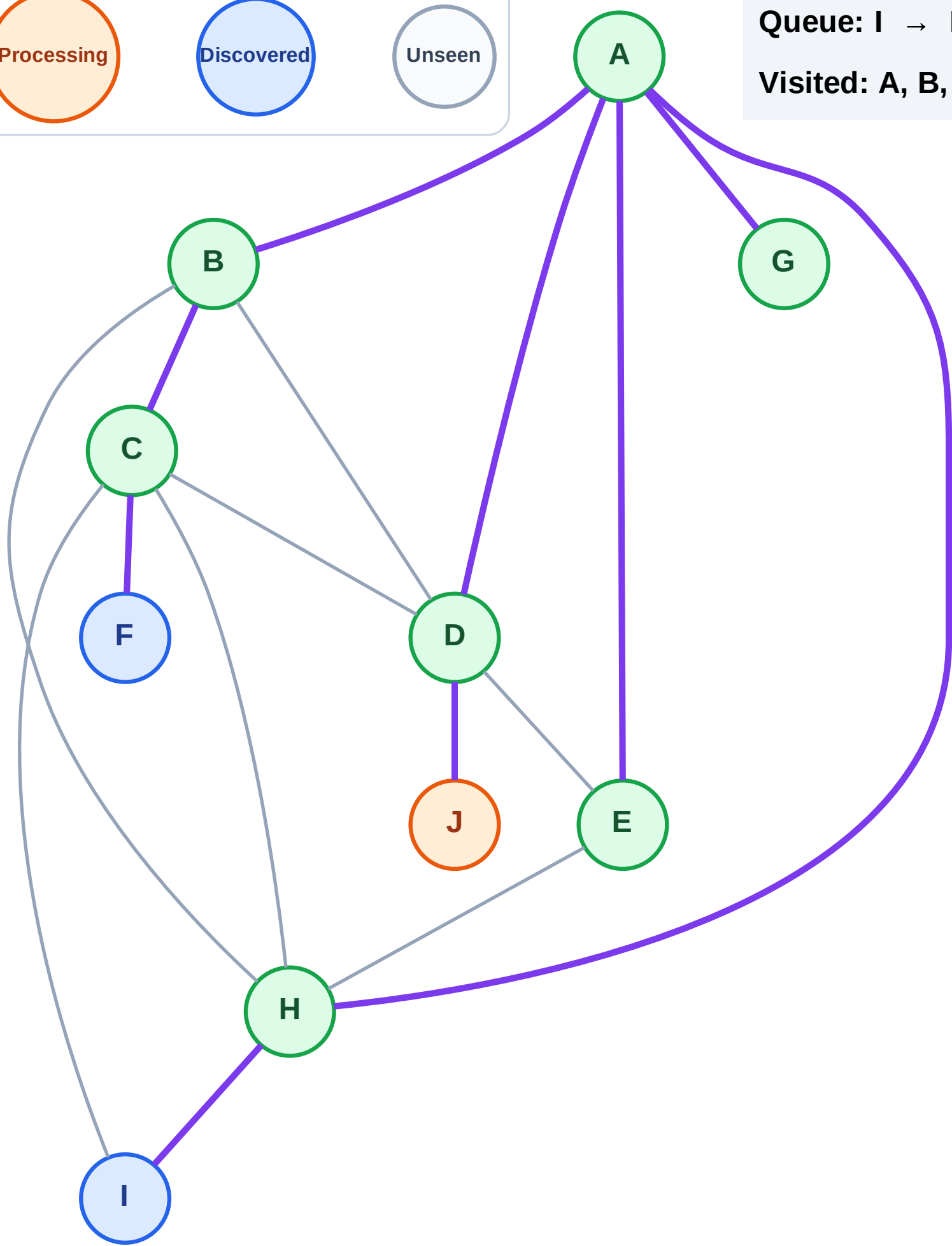
Step 16: Process node J

Legend



Queue: I → F

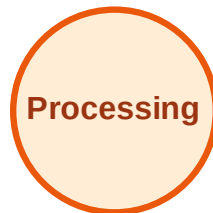
Visited: A, B, C, D, E, G, H



BFS Traversal

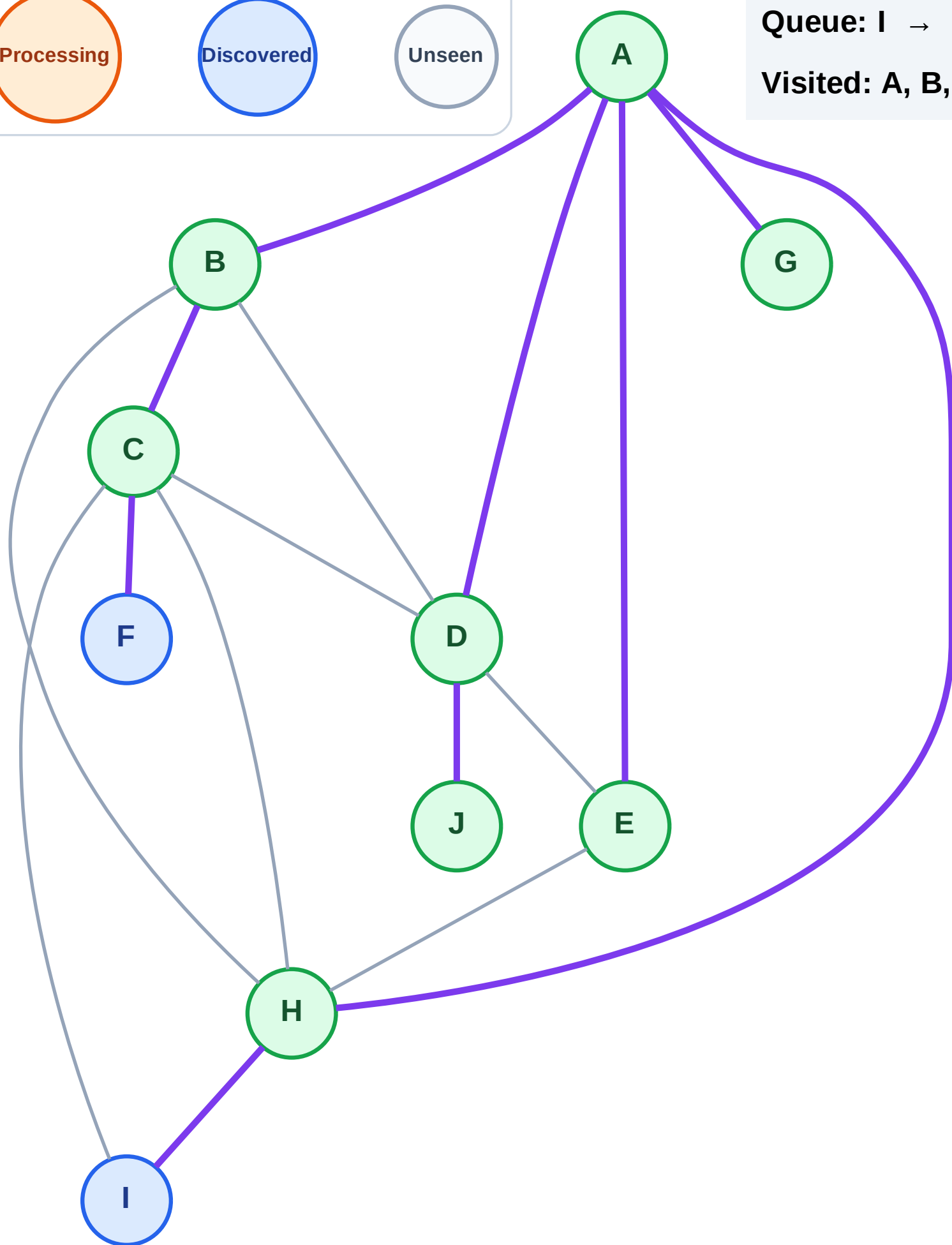
Step 17: No new neighbors from J

Legend



Queue: I → F

Visited: A, B, C, D, E, G, H, J



BFS Traversal

Step 18: Process node I

Legend

Complete

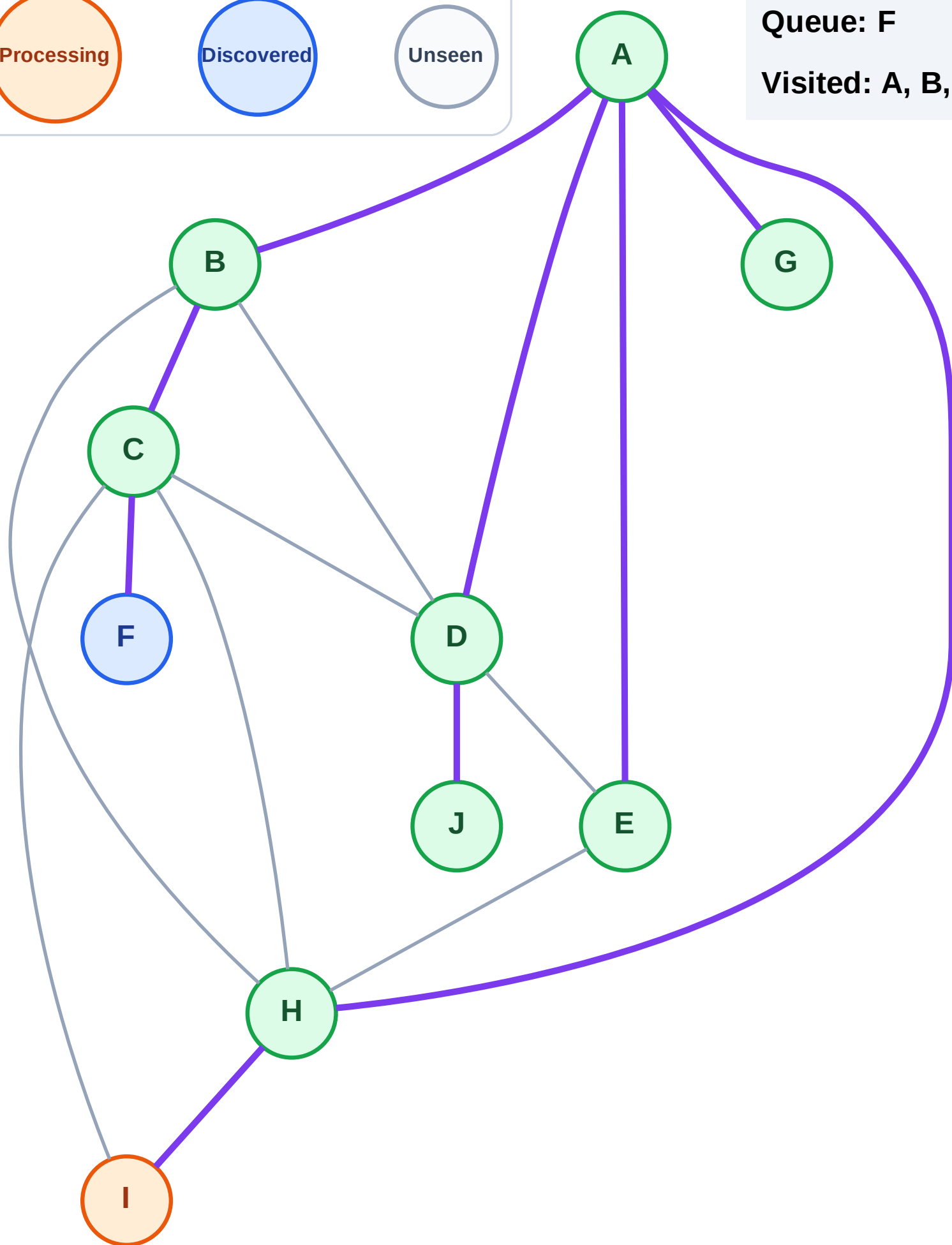
Processing

Discovered

Unseen

Queue: F

Visited: A, B, C, D, E, G, H, J



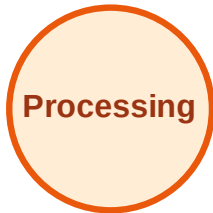
BFS Traversal

Step 19: No new neighbors from I

Legend



Complete



Processing



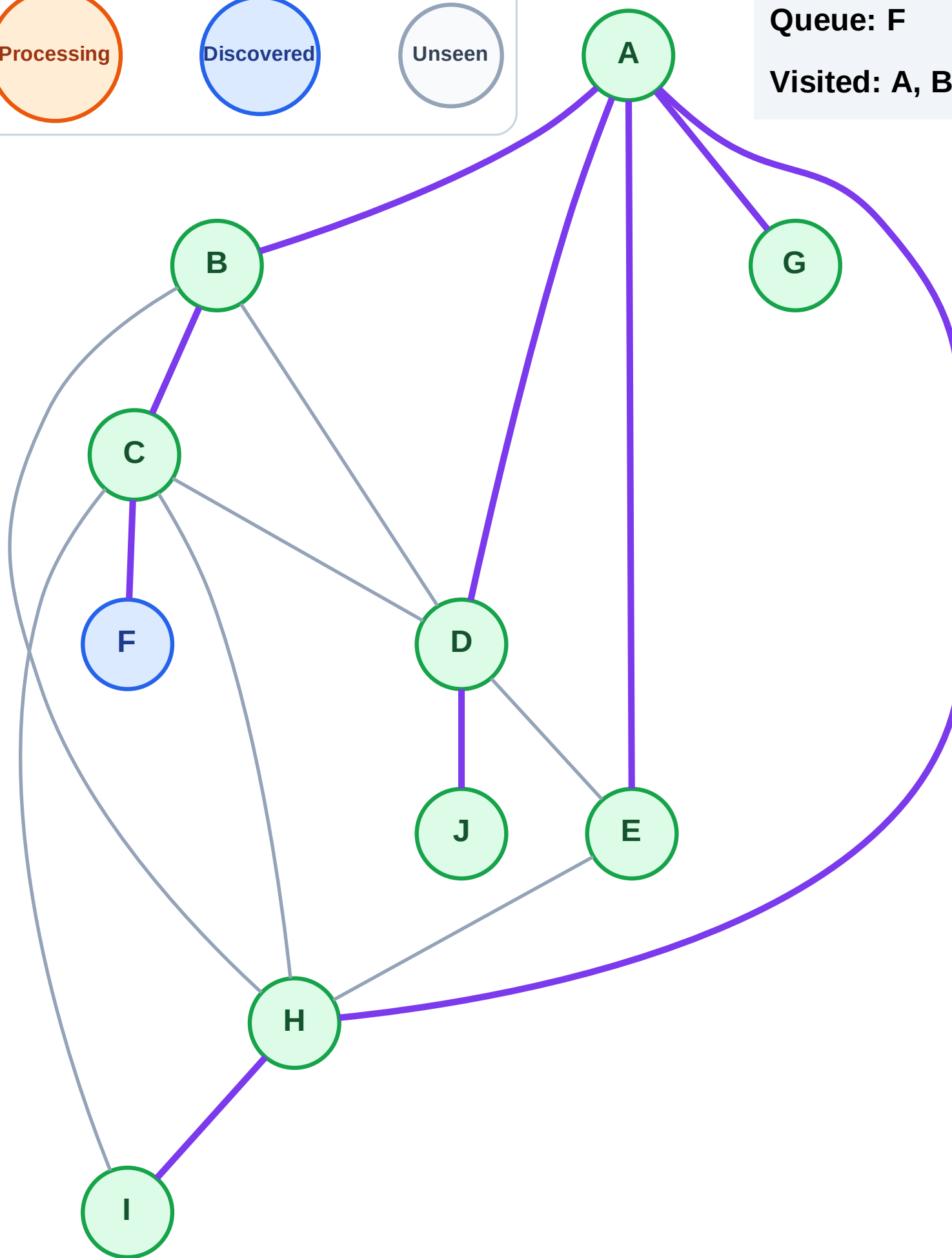
Discovered



Unseen

Queue: F

Visited: A, B, C, D, E, G, H, I, J



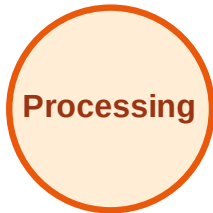
BFS Traversal

Step 20: Process node F

Legend



Complete



Processing



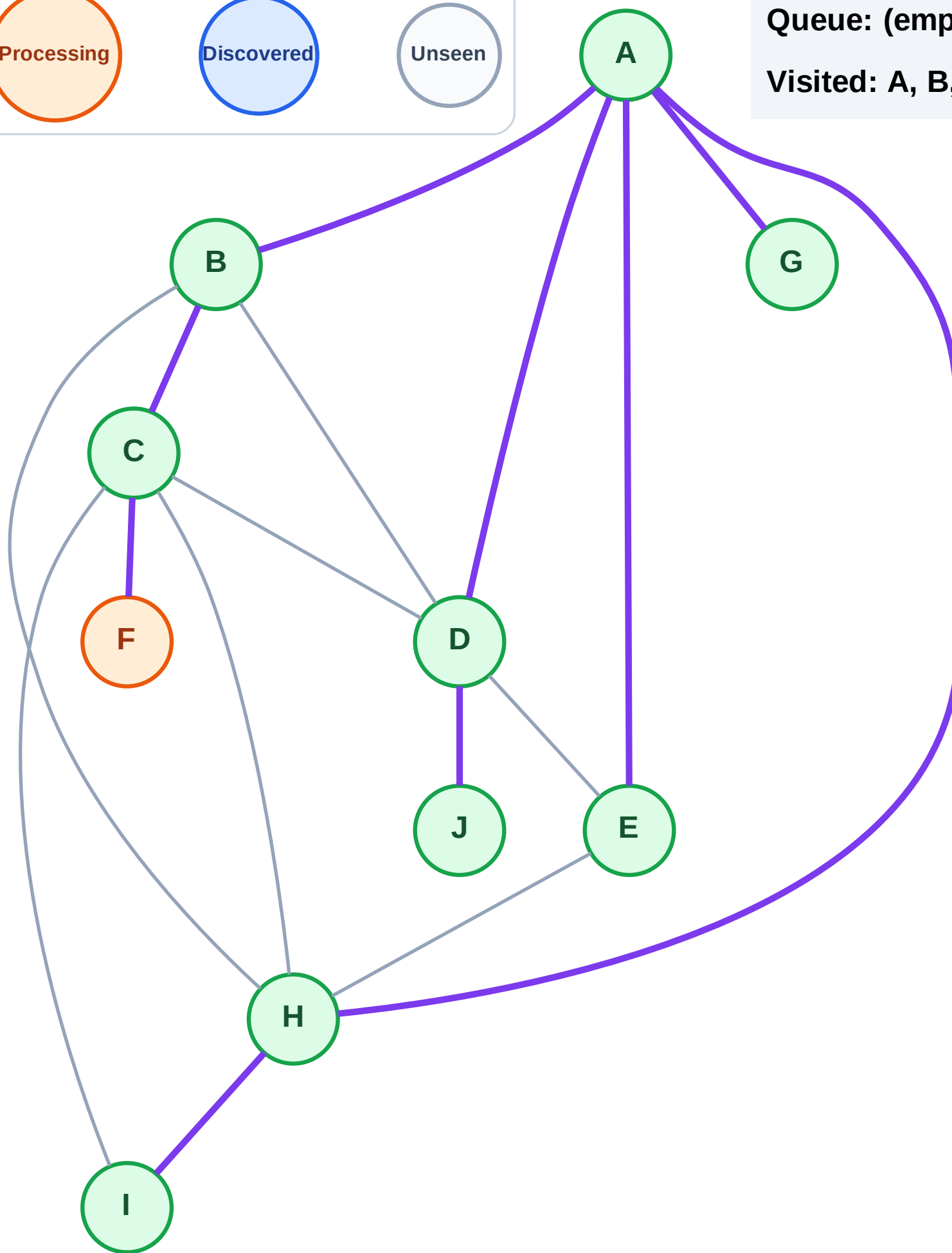
Discovered



Unseen

Queue: (empty)

Visited: A, B, C, D, E, G, H, I, J



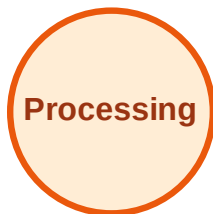
BFS Traversal

Step 21: No new neighbors from F

Legend



Complete



Processing



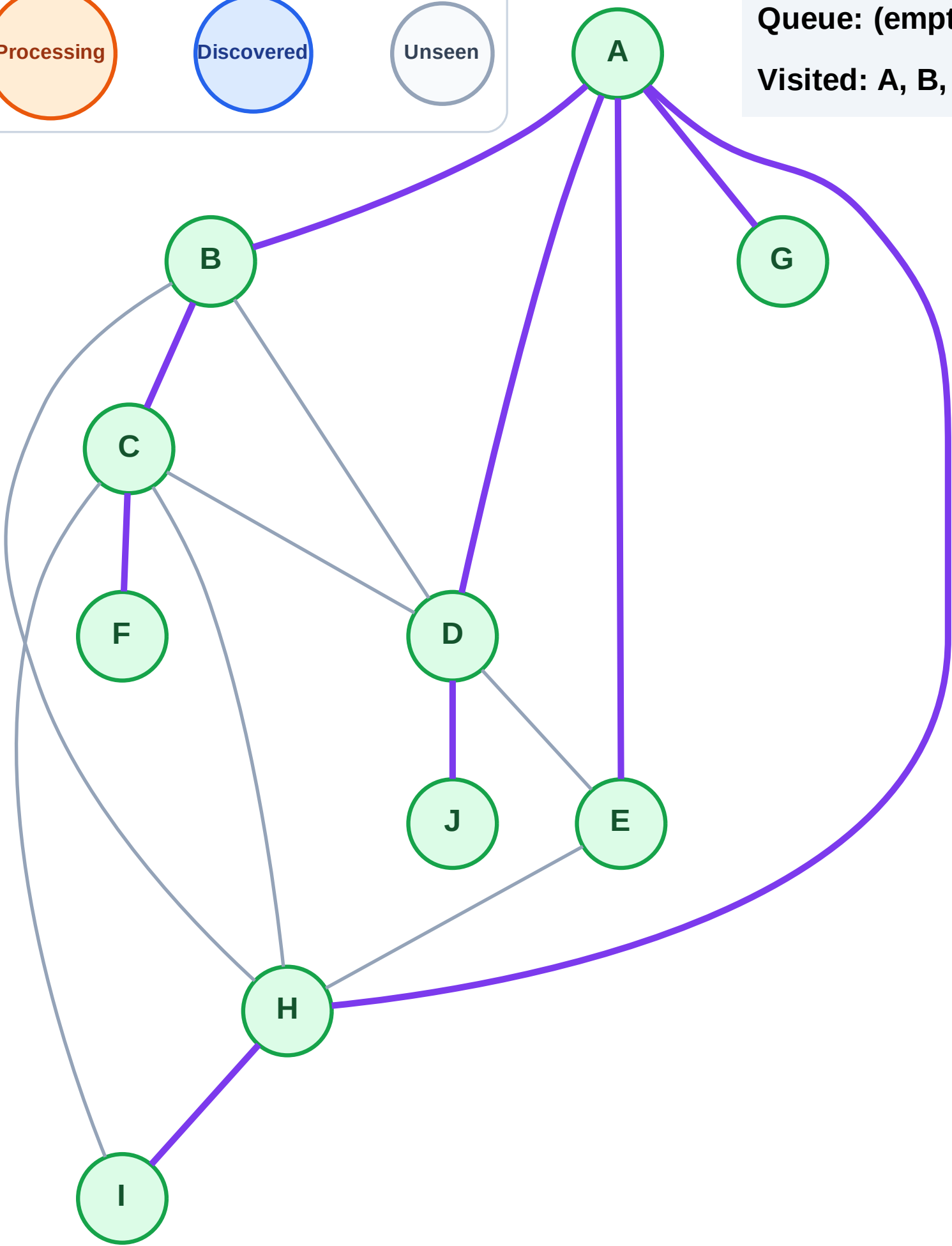
Discovered



Unseen

Queue: (empty)

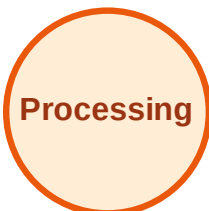
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

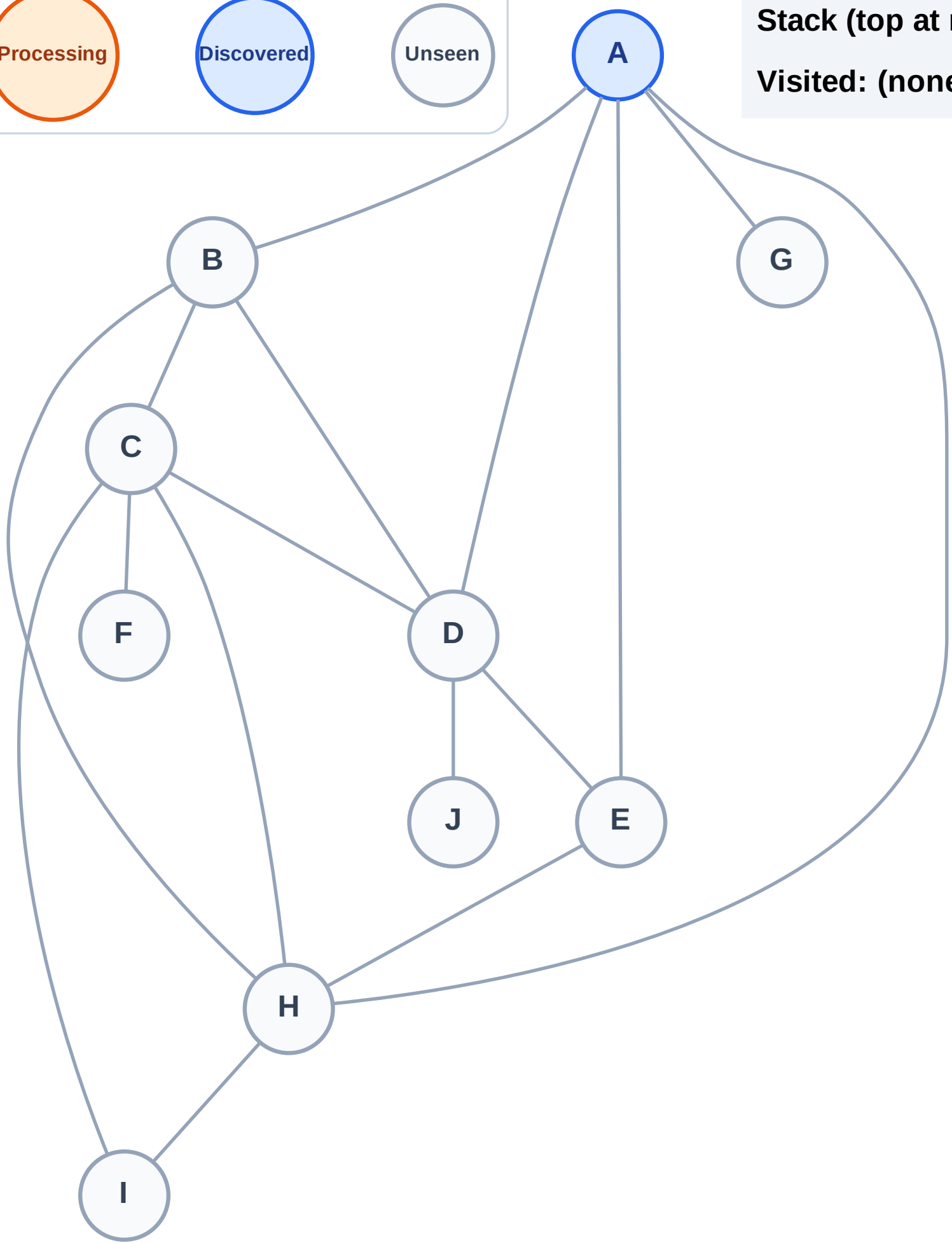
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

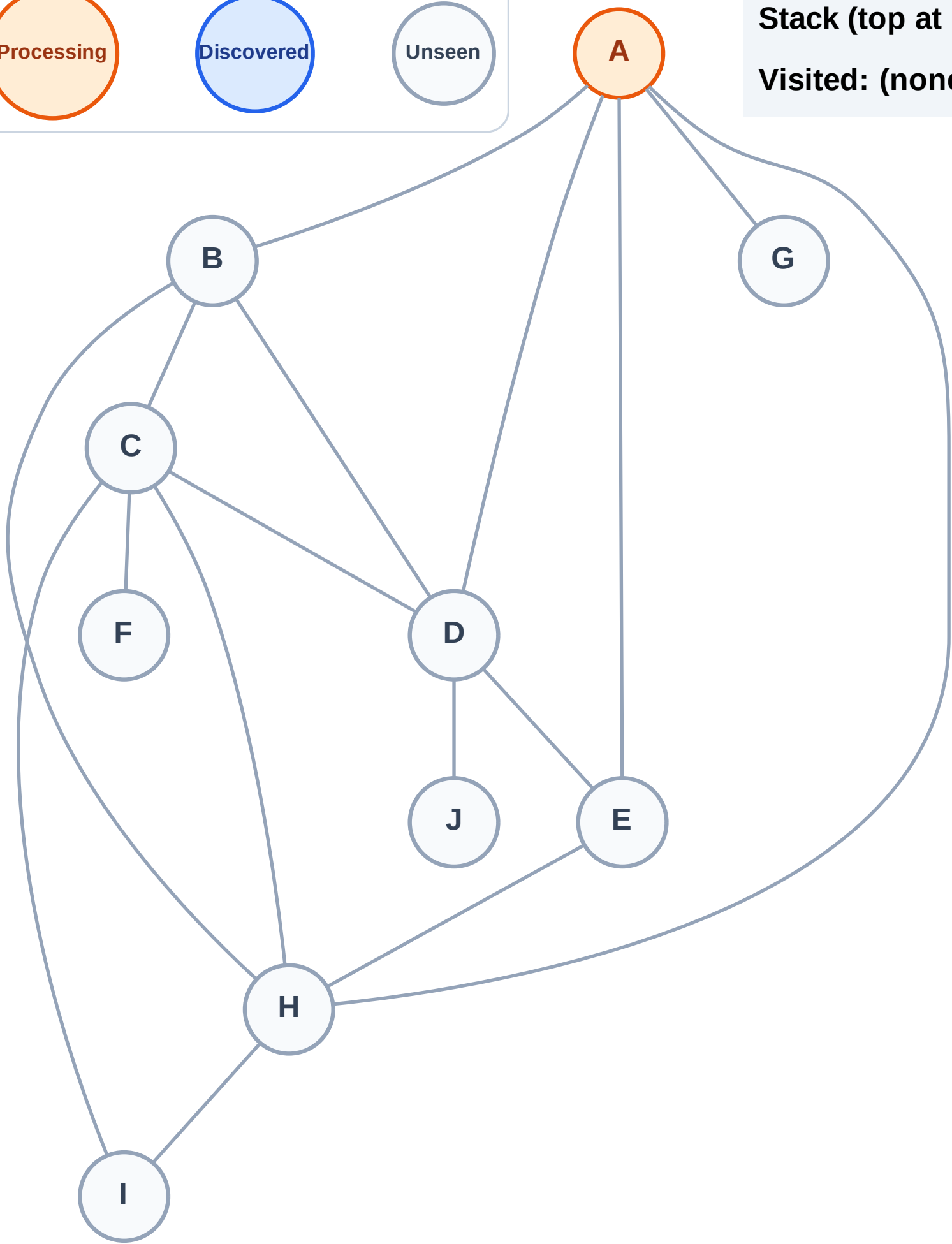
Complete

Processing

Discovered

Unseen

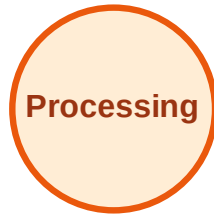
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

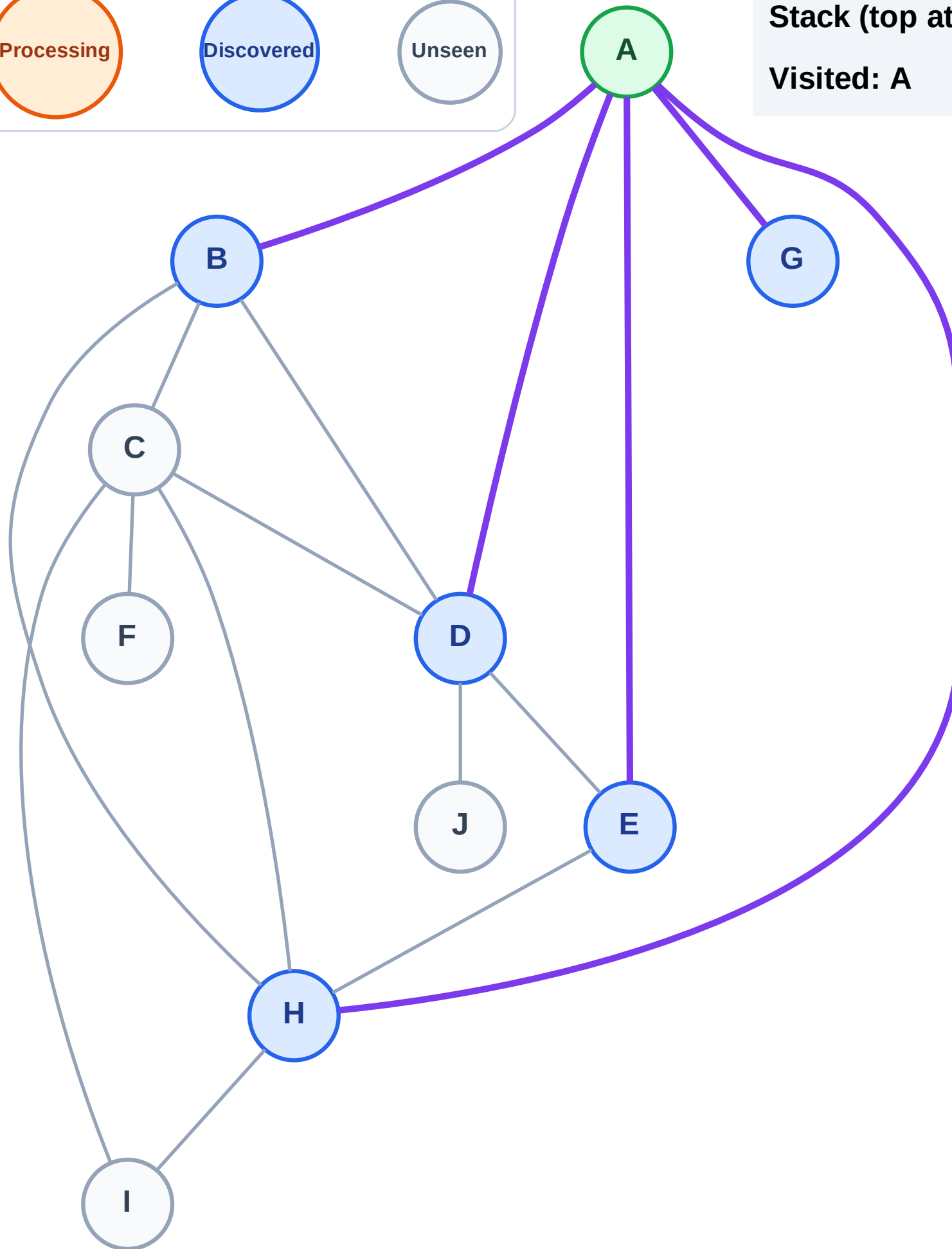
Step 3: Discover H, G, E, D, B from A

Legend



Stack (top at right): H | G | E | D | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

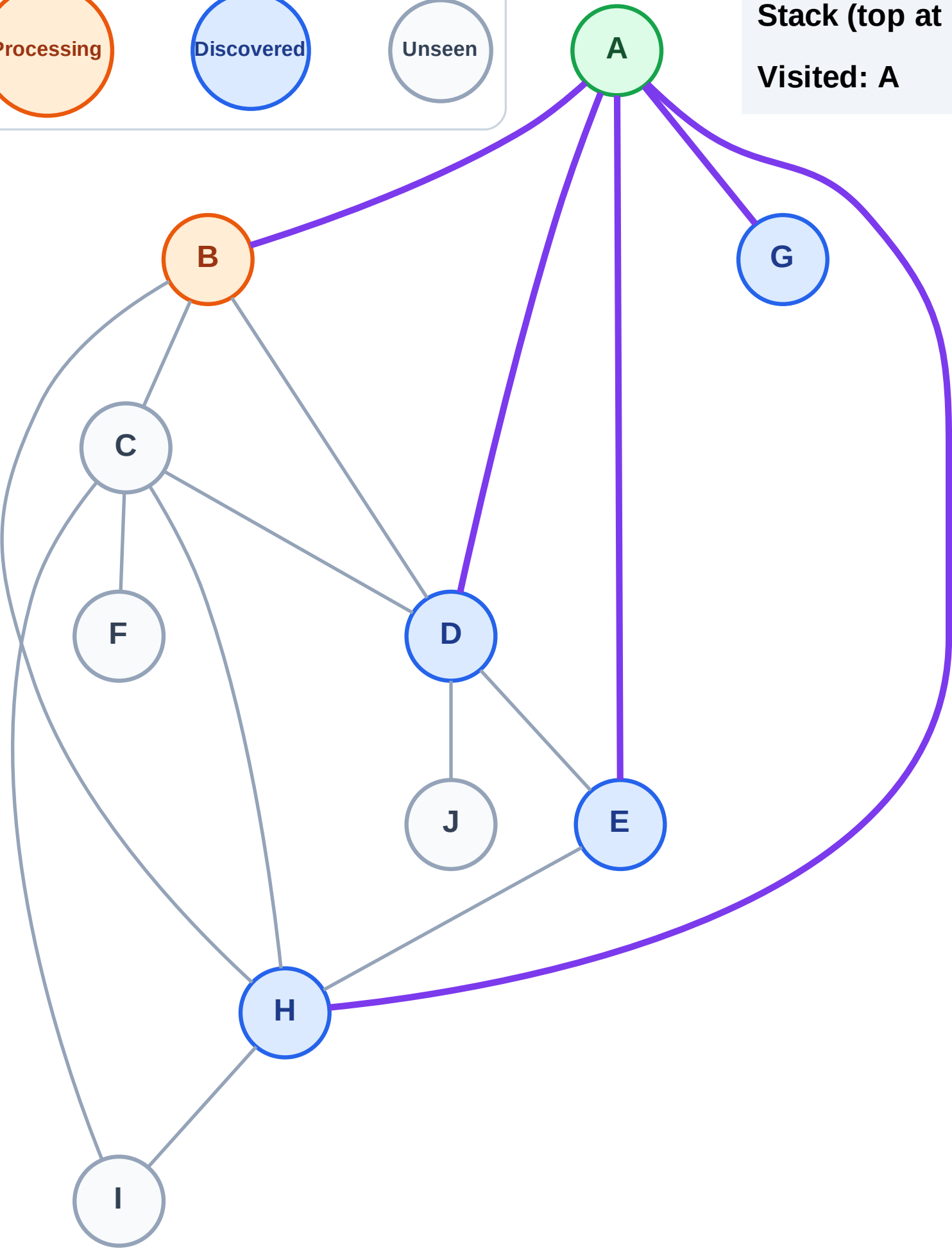
Processing

Discovered

Unseen

Stack (top at right): H | G | E | D

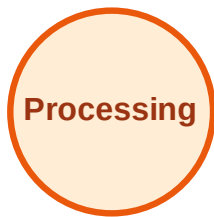
Visited: A



DFS Traversal

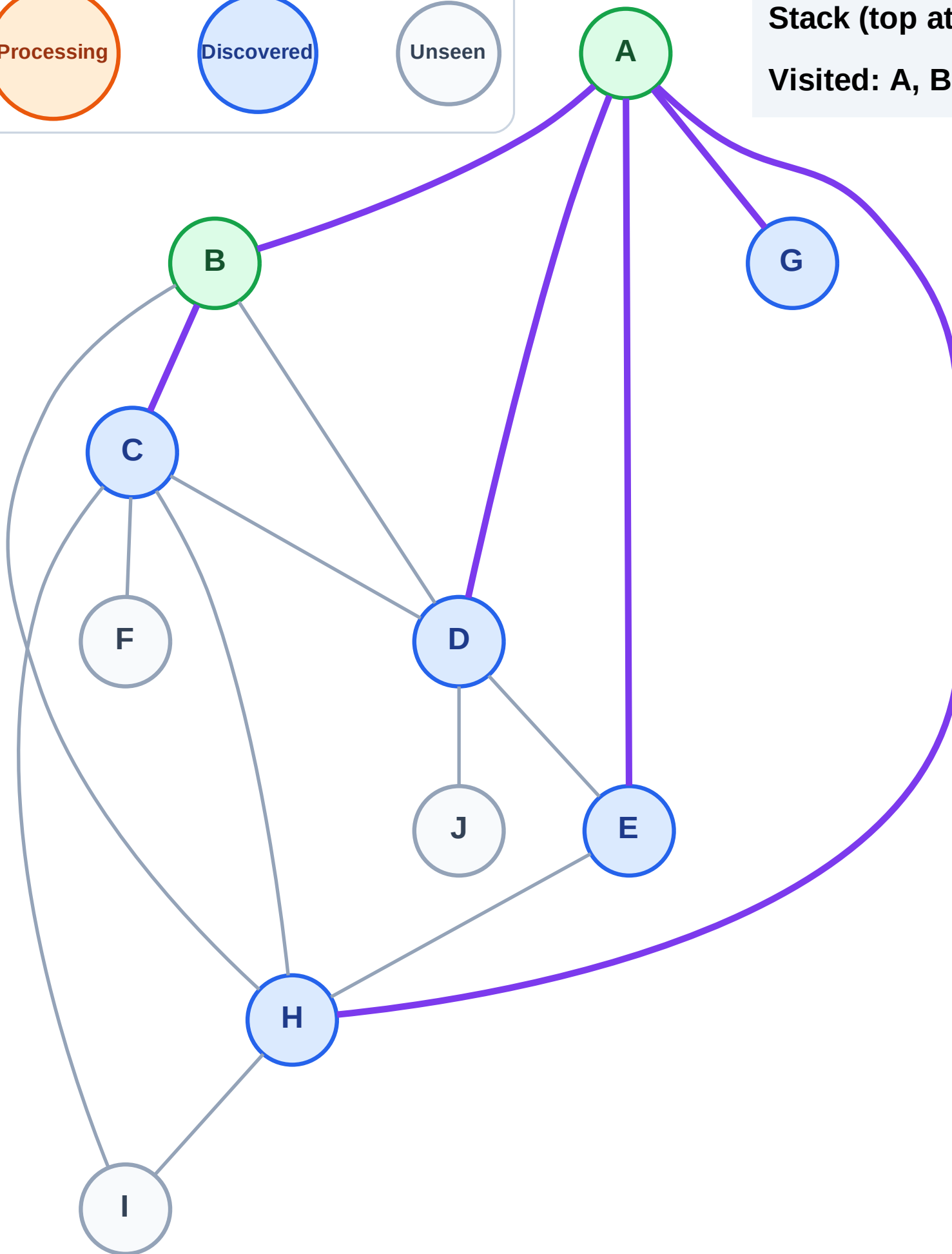
Step 5: Discover C from B

Legend



Stack (top at right): H | G | E | D | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

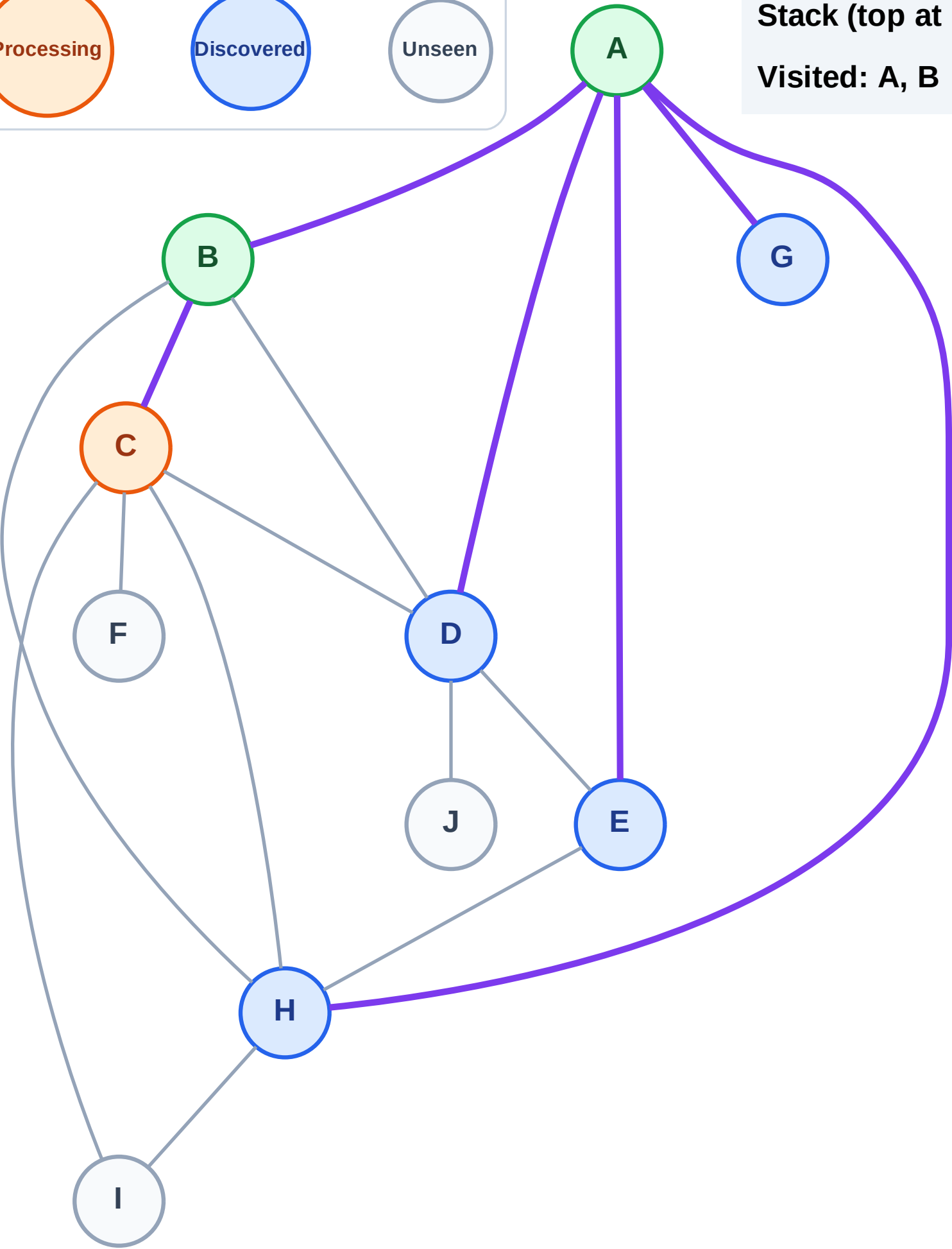
Complete

Processing

Discovered

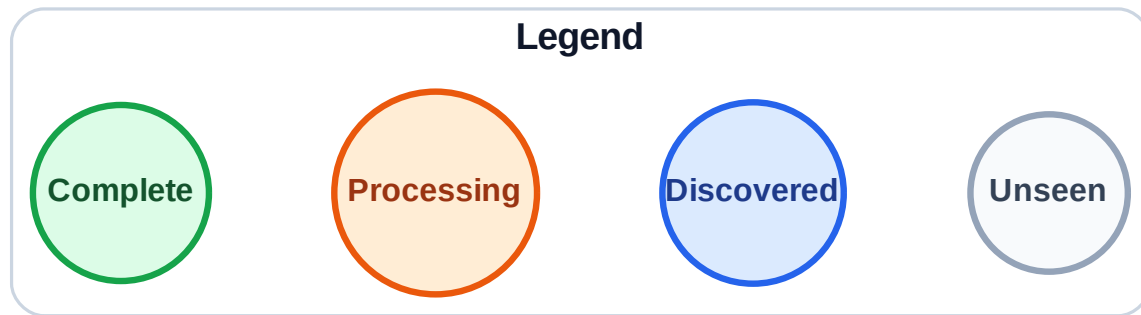
Unseen

Stack (top at right): H | G | E | D
Visited: A, B



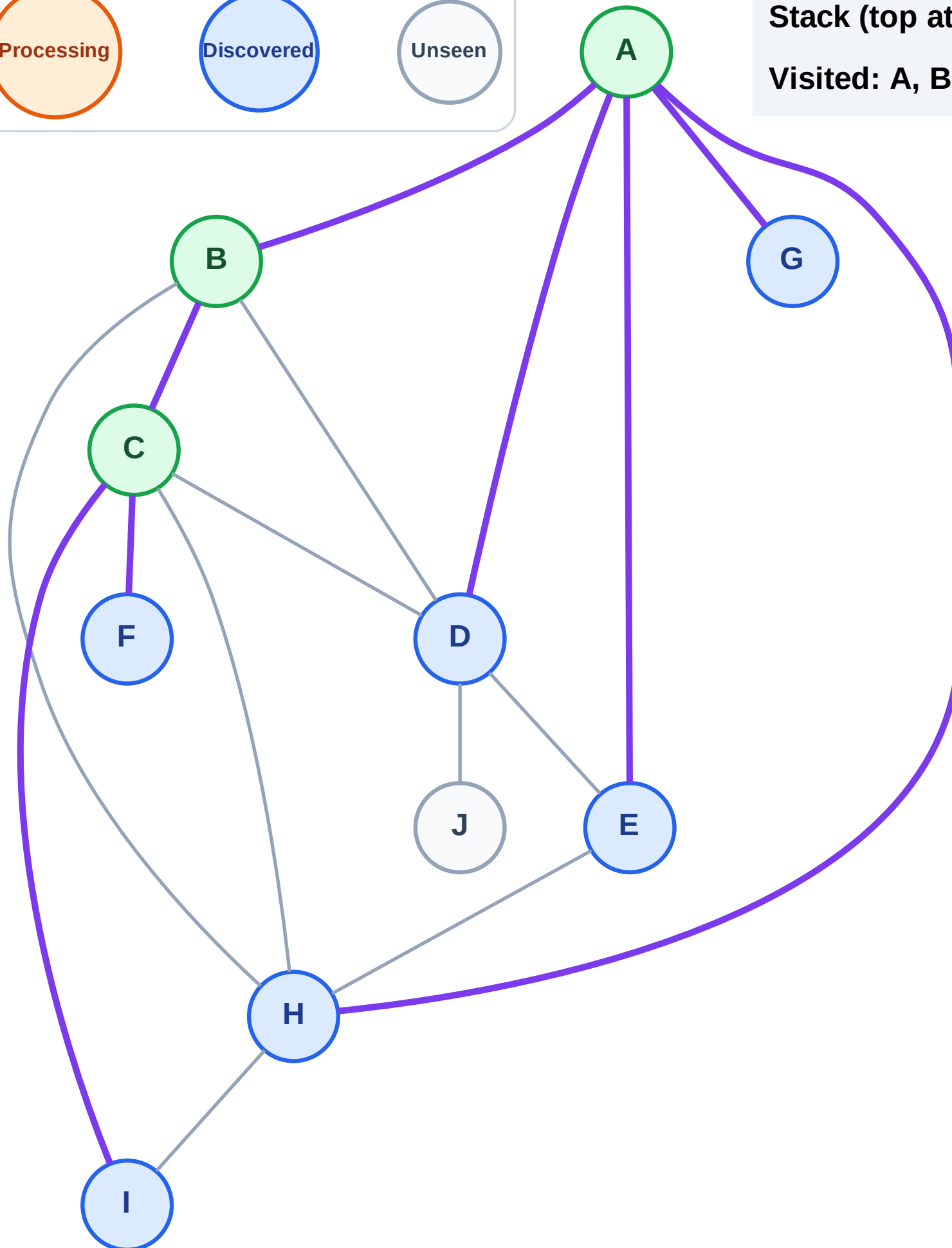
Step 7: Discover I, F from C

Step 7: Discover I, F from C



Stack (top at right): H | G | E | D | I | F

Visited: A, B, C



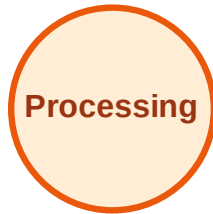
DFS Traversal

Step 8: Process node F

Legend



Complete



Processing



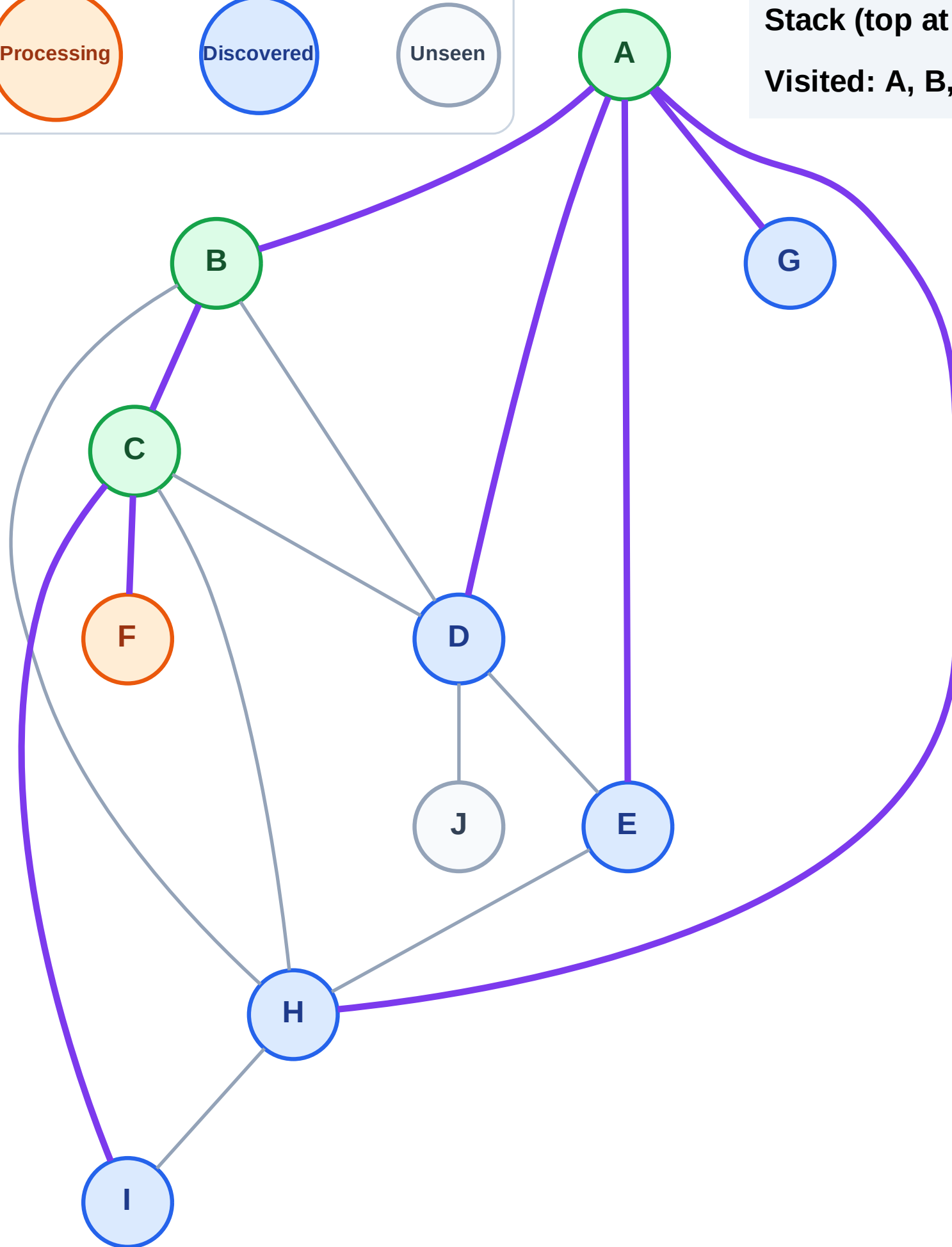
Discovered



Unseen

Stack (top at right): H | G | E | D | I

Visited: A, B, C



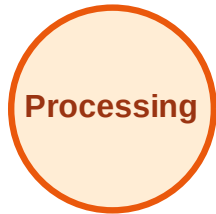
DFS Traversal

Step 9: No new neighbors from F

Legend



Complete



Processing



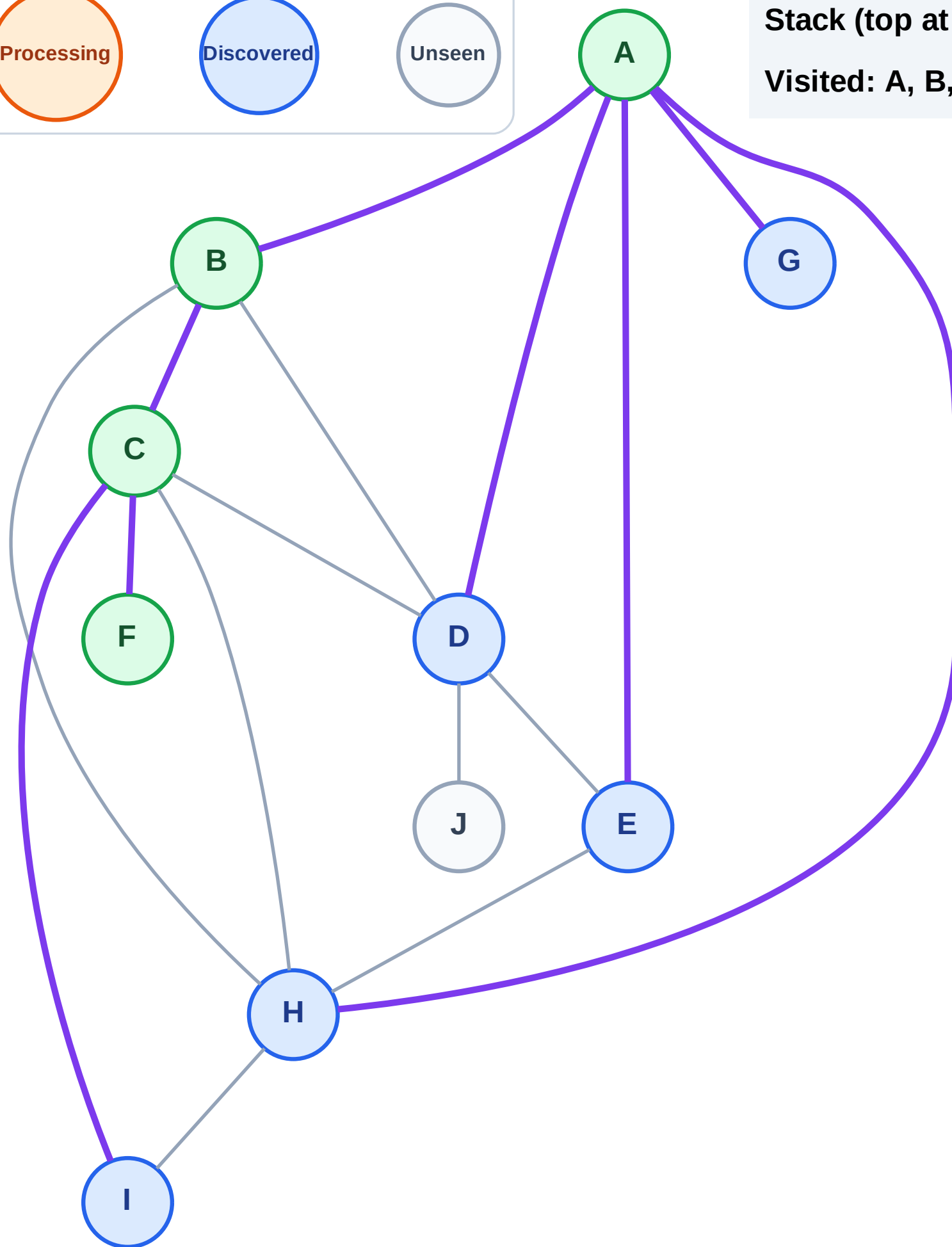
Discovered



Unseen

Stack (top at right): H | G | E | D | I

Visited: A, B, C, F



DFS Traversal

Step 10: Process node I

Legend

Complete

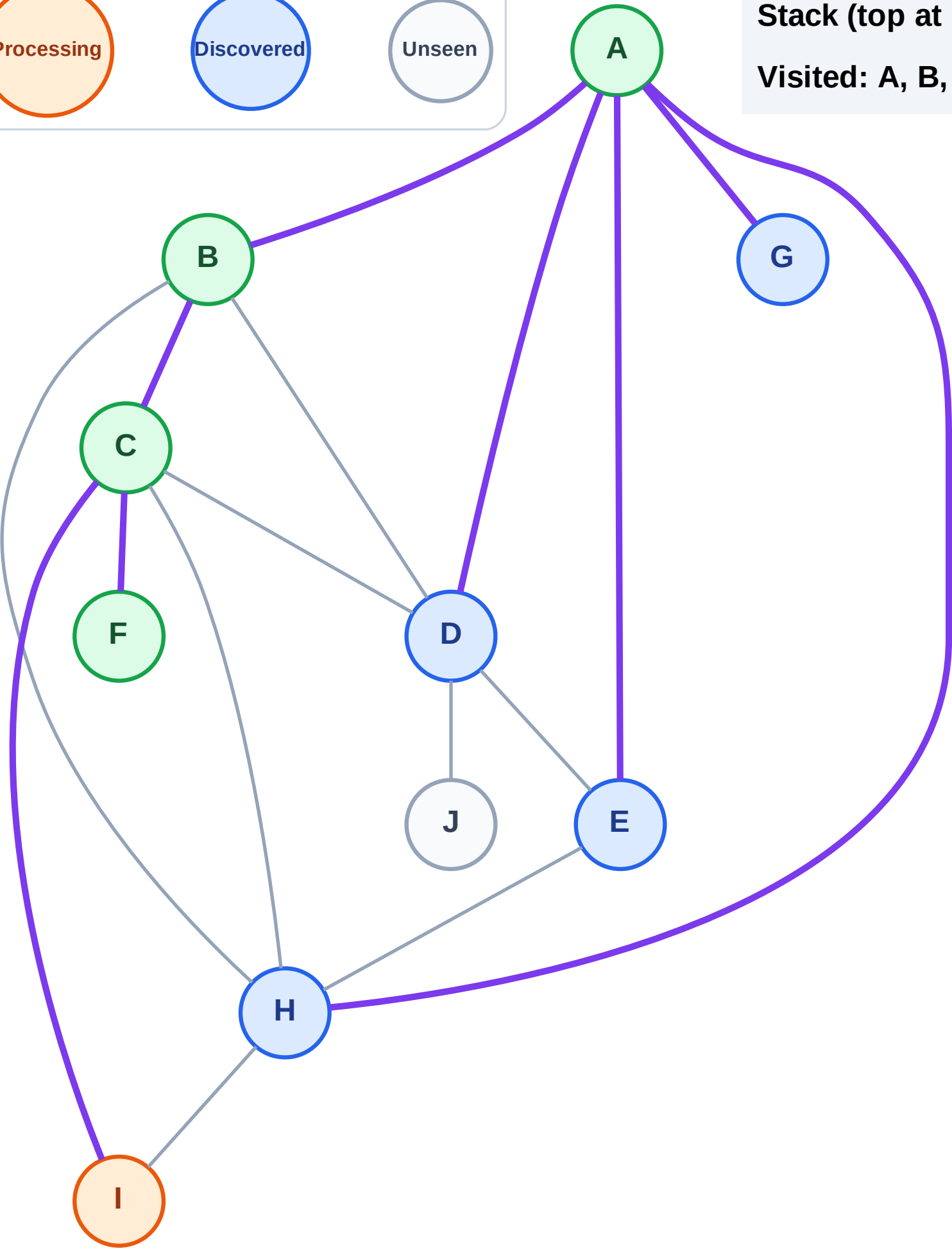
Processing

Discovered

Unseen

Stack (top at right): H | G | E | D

Visited: A, B, C, F



DFS Traversal

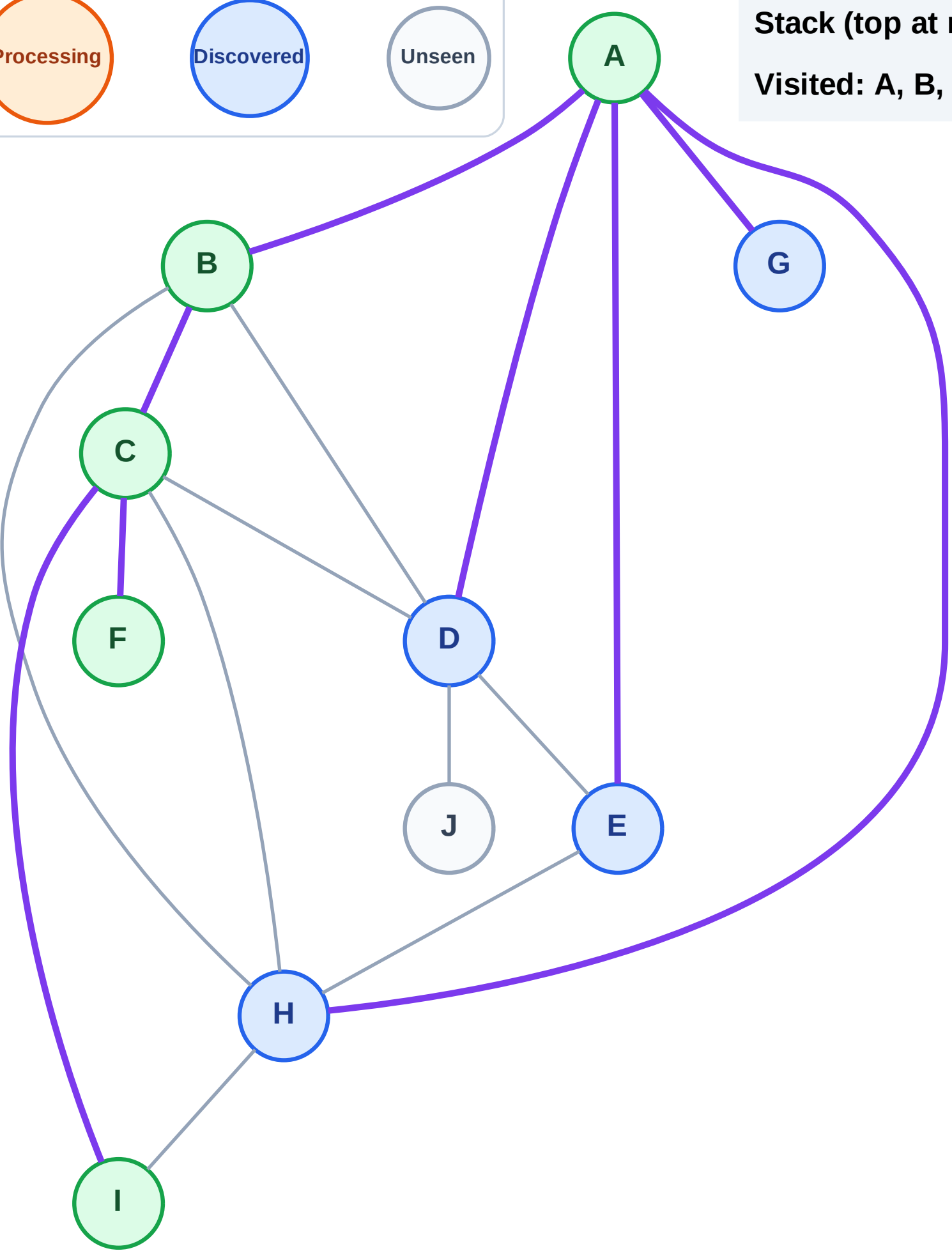
Step 11: No new neighbors from I

Legend

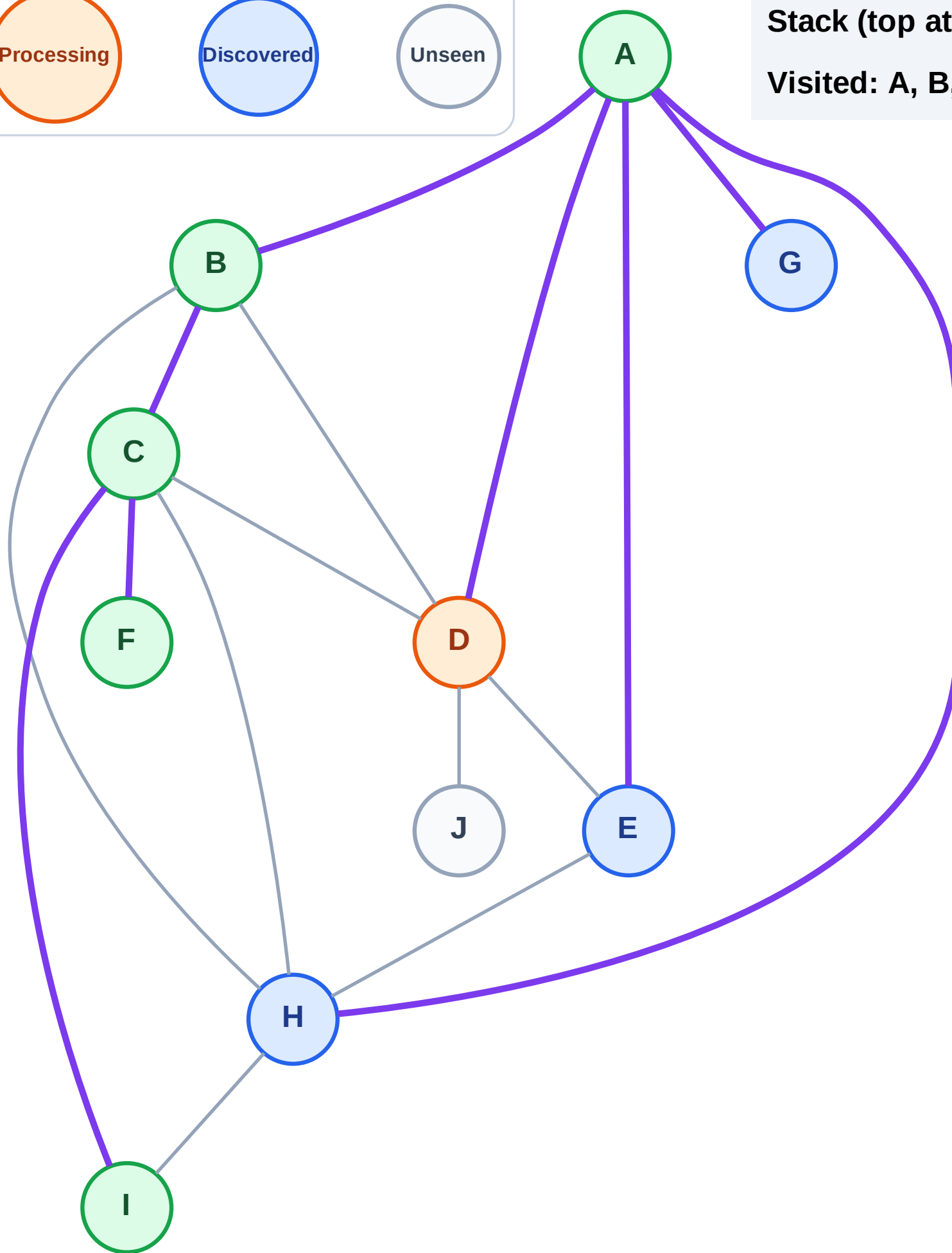
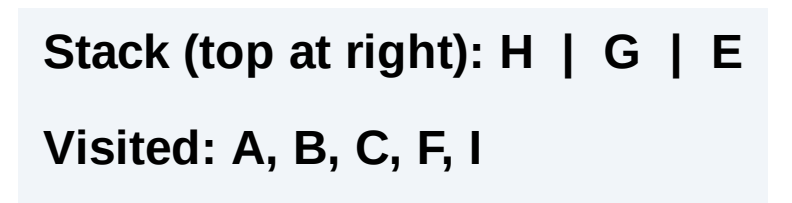


Stack (top at right): H | G | E | D

Visited: A, B, C, F, I



Step 12: Process node D



DFS Traversal

Step 13: Discover J from D

Legend

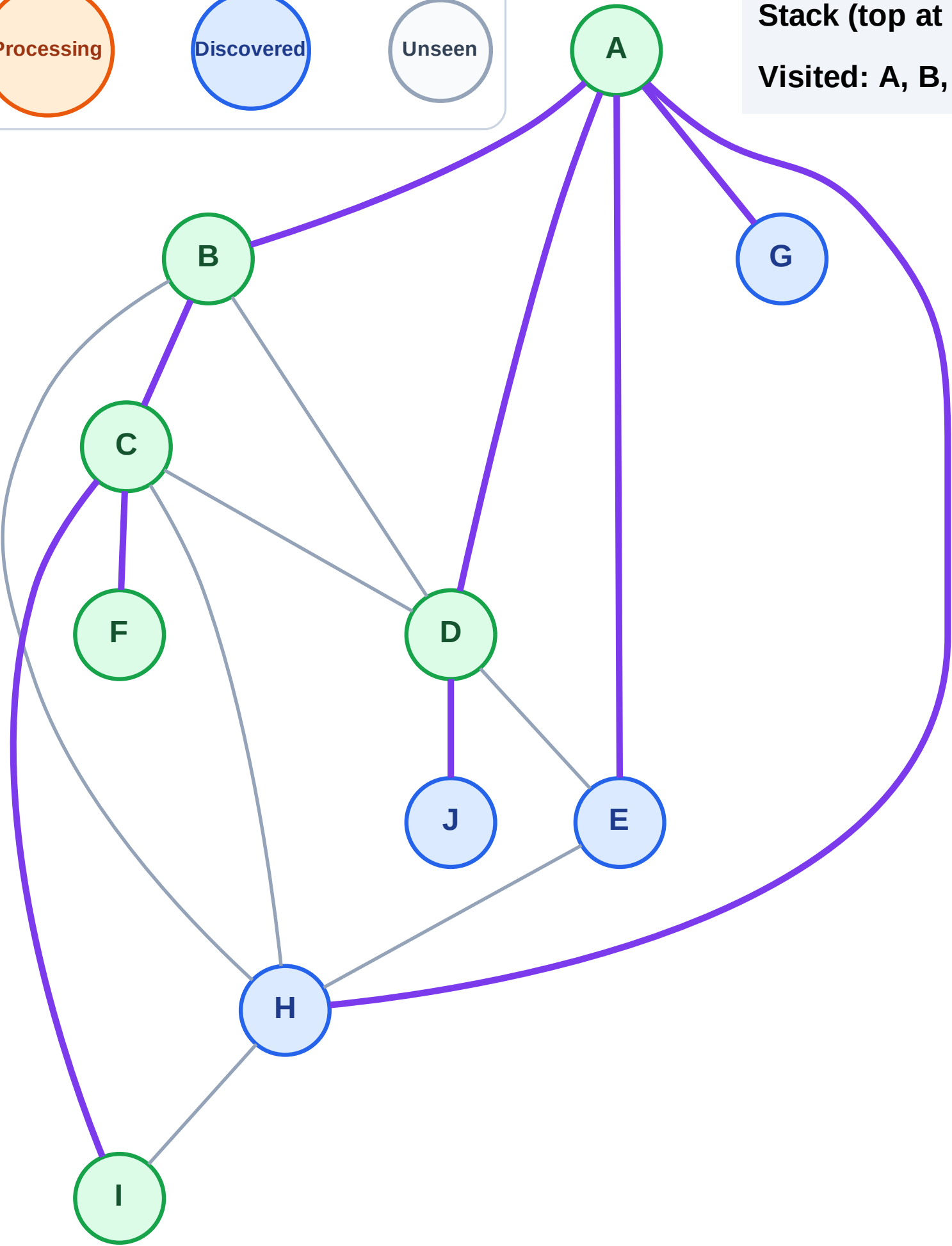
Complete

Processing

Discovered

Unseen

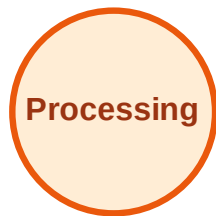
Stack (top at right): H | G | E | J
Visited: A, B, C, D, F, I



DFS Traversal

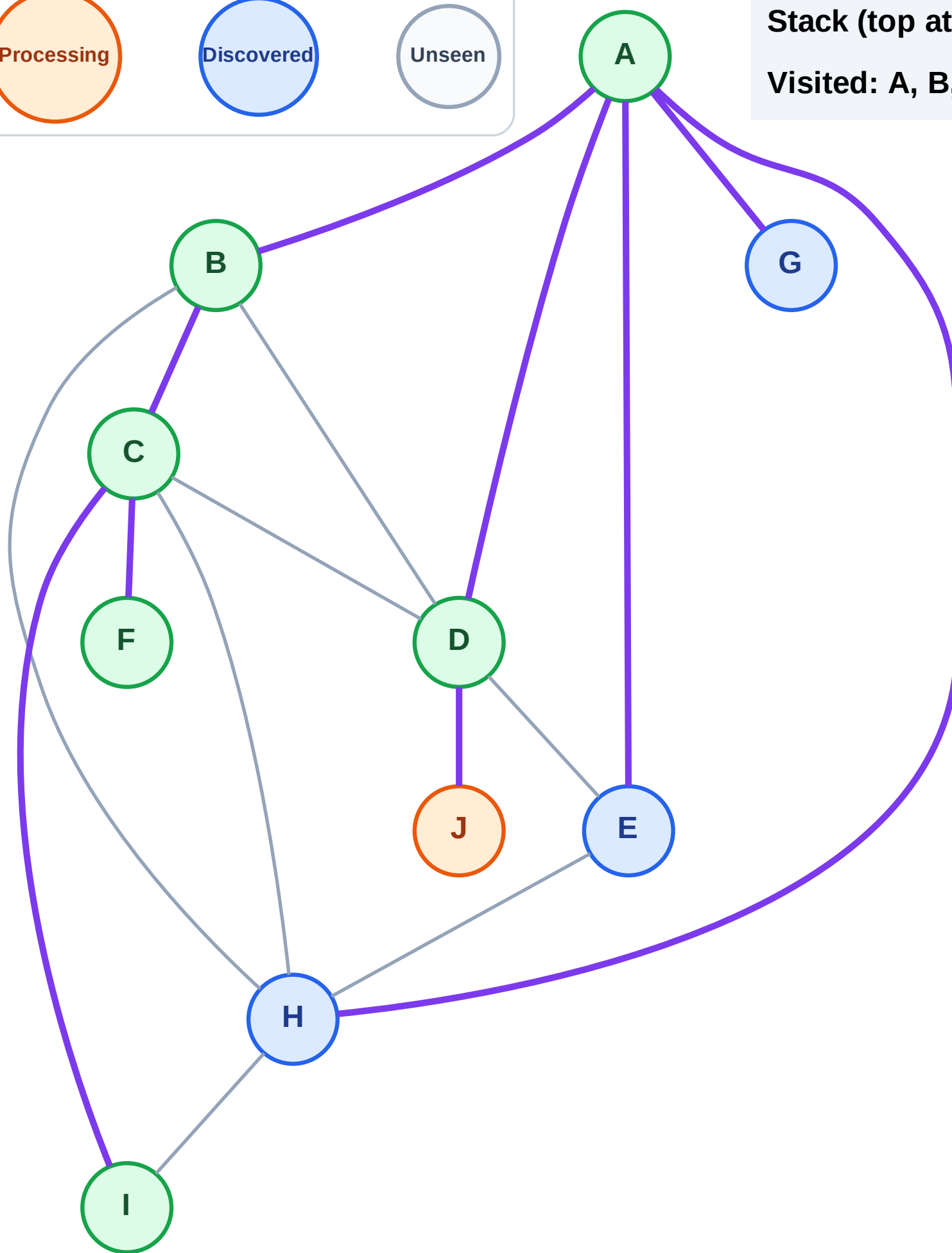
Step 14: Process node J

Legend



Stack (top at right): H | G | E

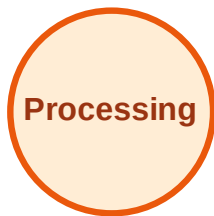
Visited: A, B, C, D, F, I



DFS Traversal

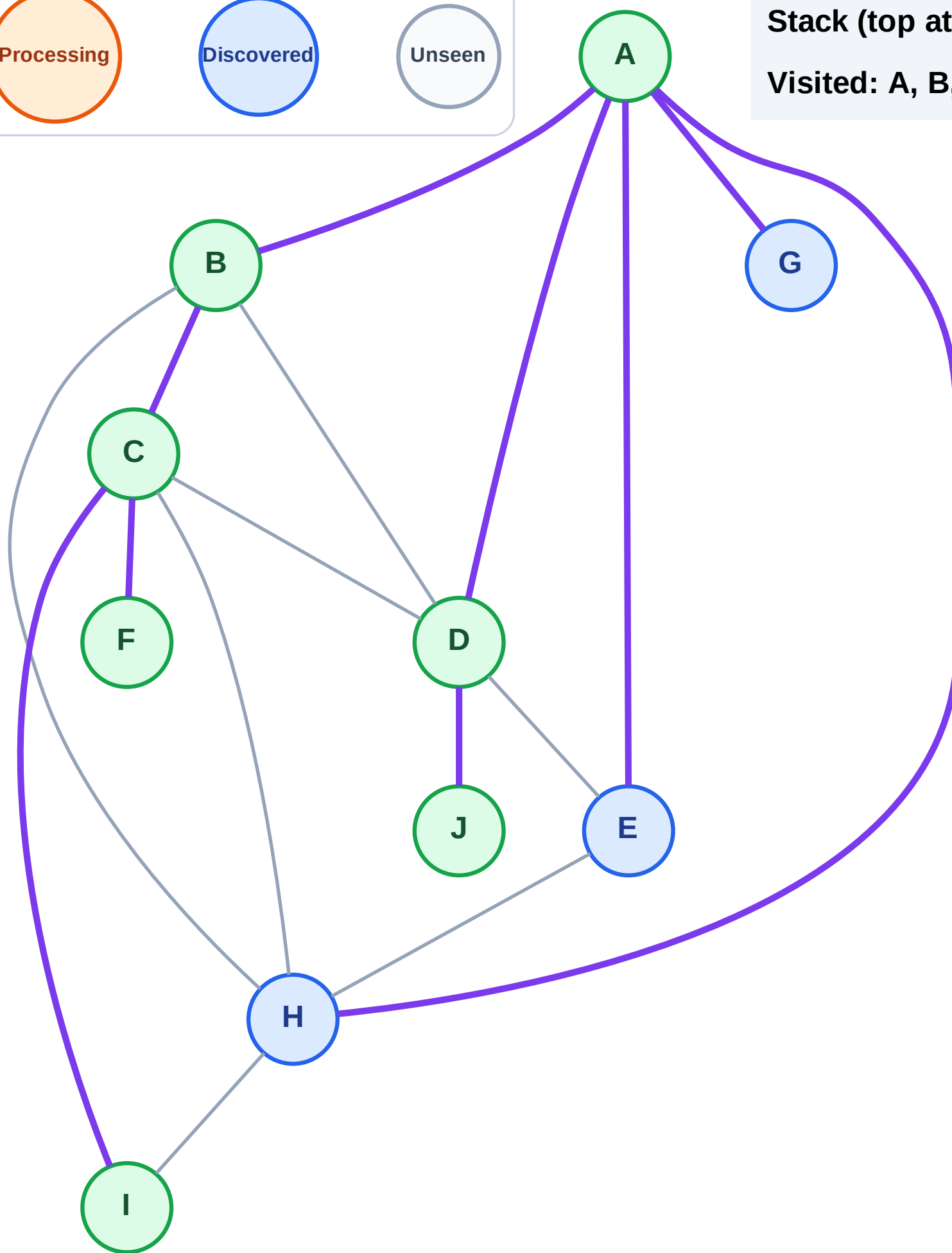
Step 15: No new neighbors from J

Legend



Stack (top at right): H | G | E

Visited: A, B, C, D, F, I, J



DFS Traversal

Step 16: Process node E

Legend

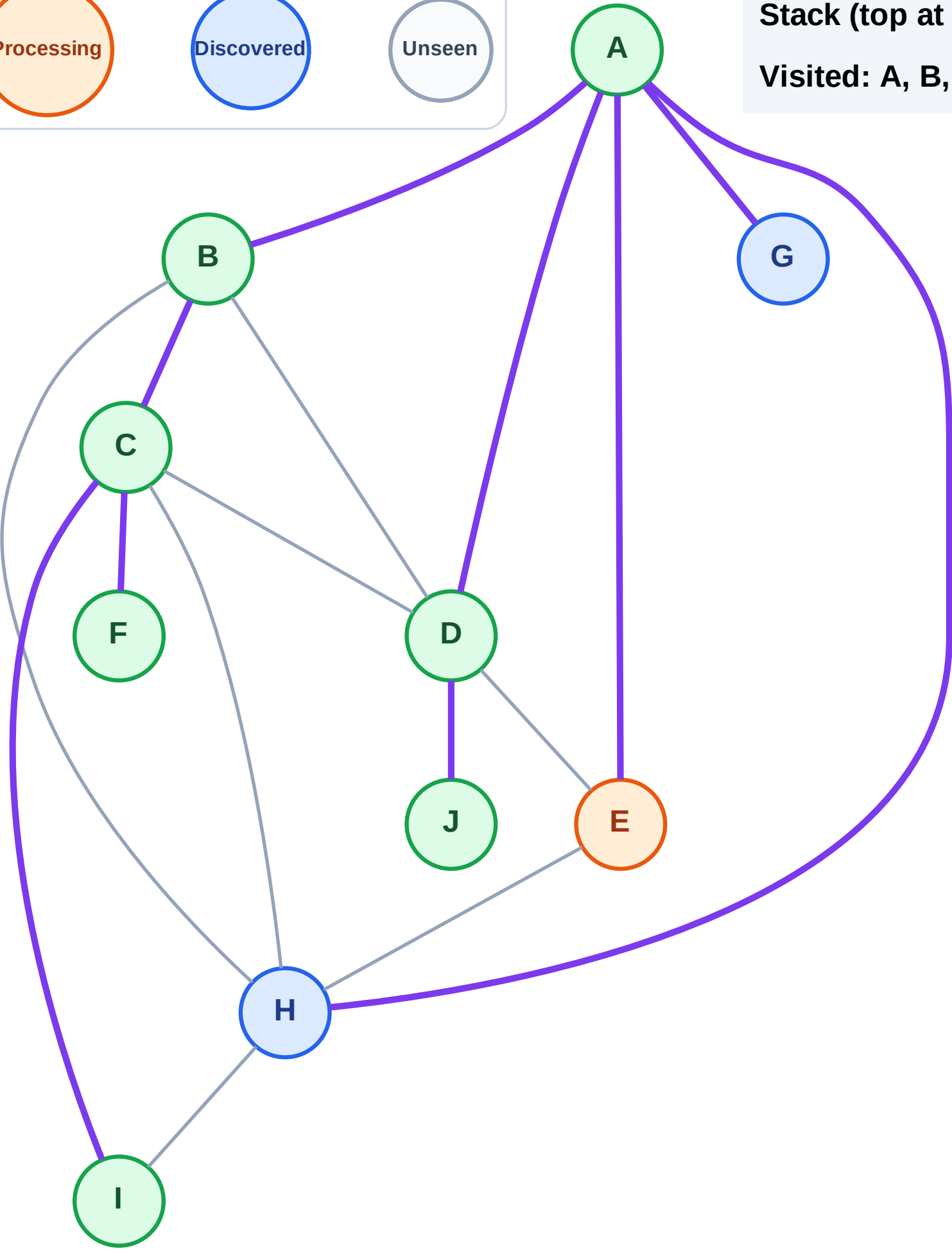
Complete

Processing

Discovered

Unseen

Stack (top at right): H | G
Visited: A, B, C, D, F, I, J



DFS Traversal

Step 17: No new neighbors from E

Legend

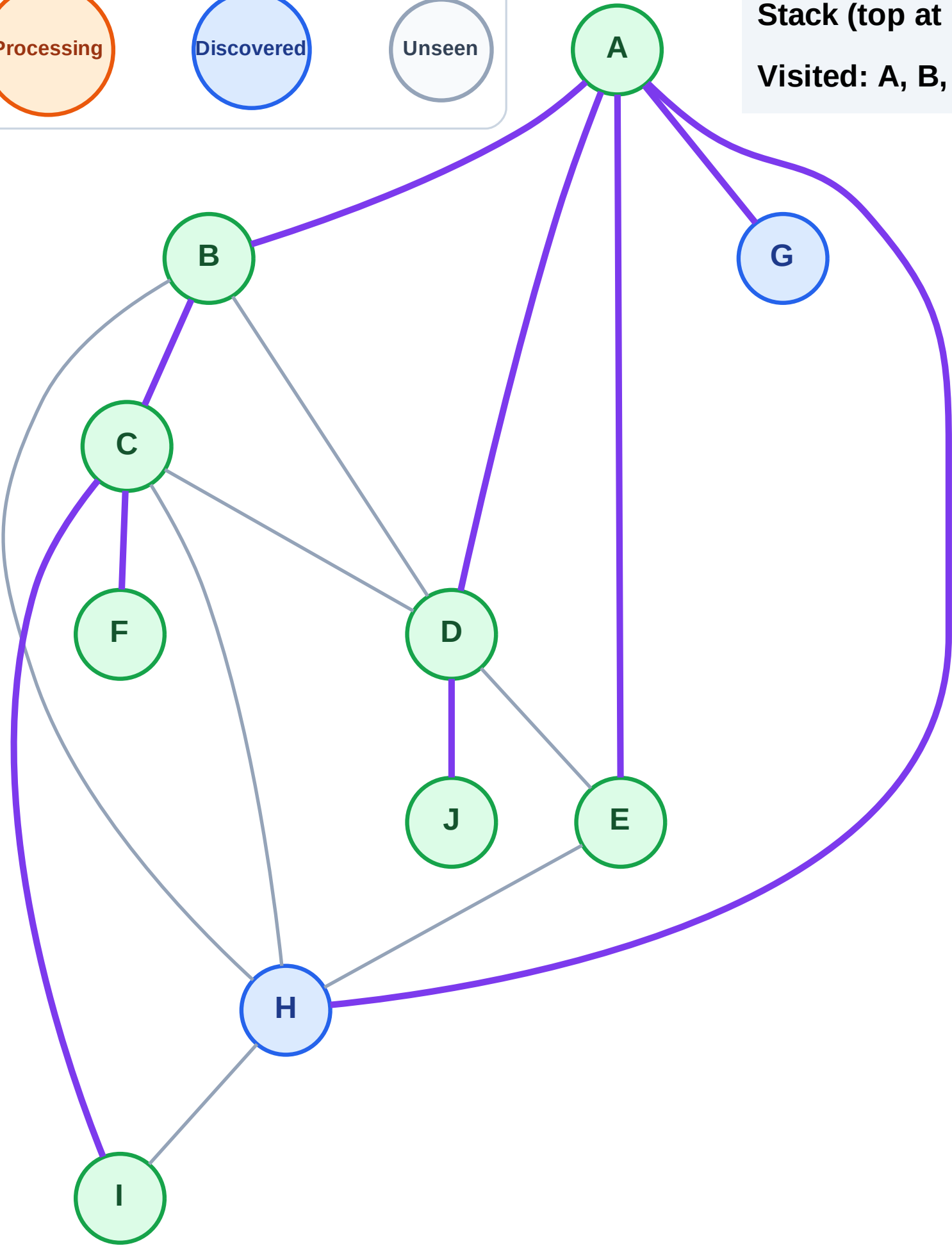
Complete

Processing

Discovered

Unseen

Stack (top at right): H | G
Visited: A, B, C, D, E, F, I, J



DFS Traversal

Step 18: Process node G

Legend

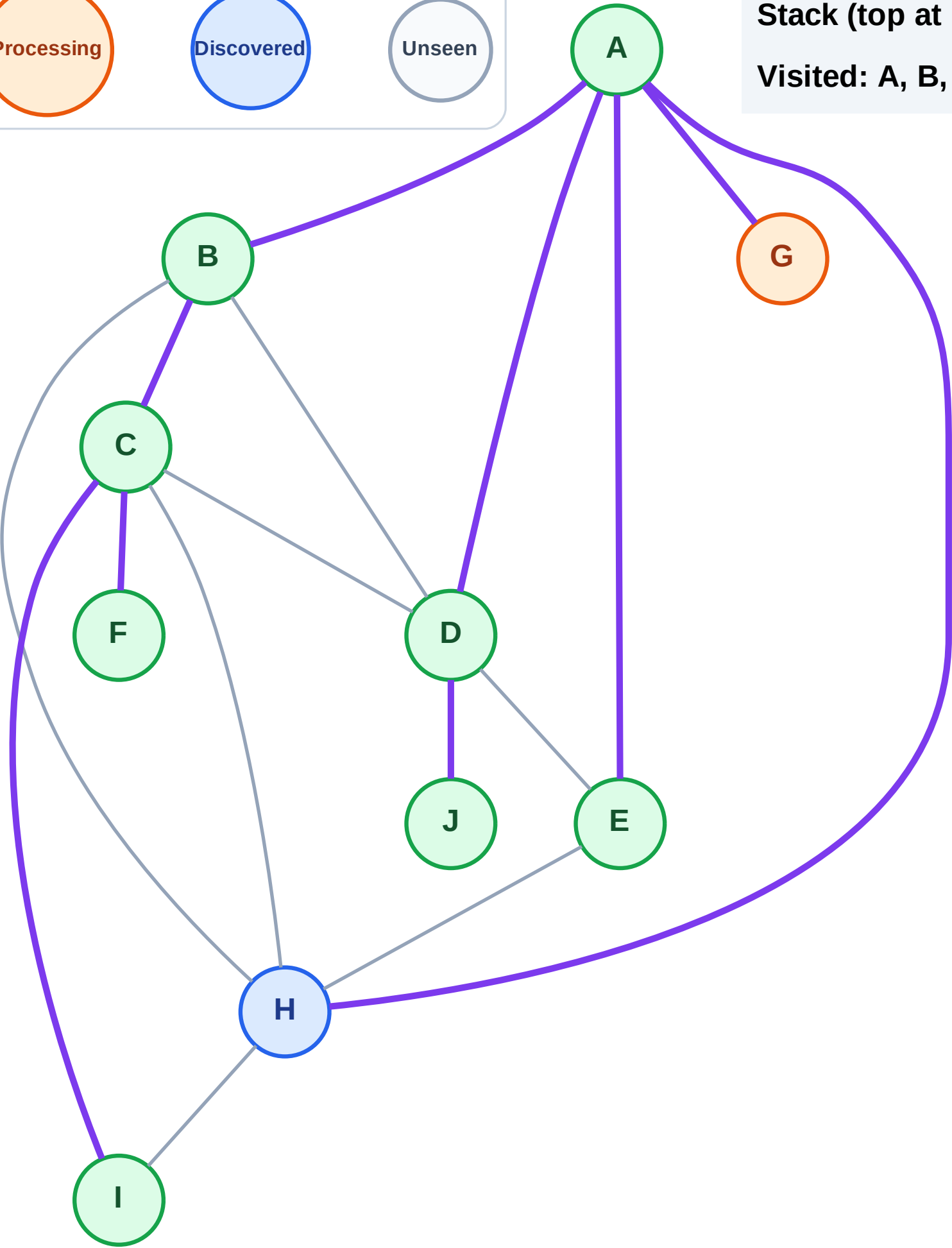
Complete

Processing

Discovered

Unseen

Stack (top at right): H
Visited: A, B, C, D, E, F, I, J

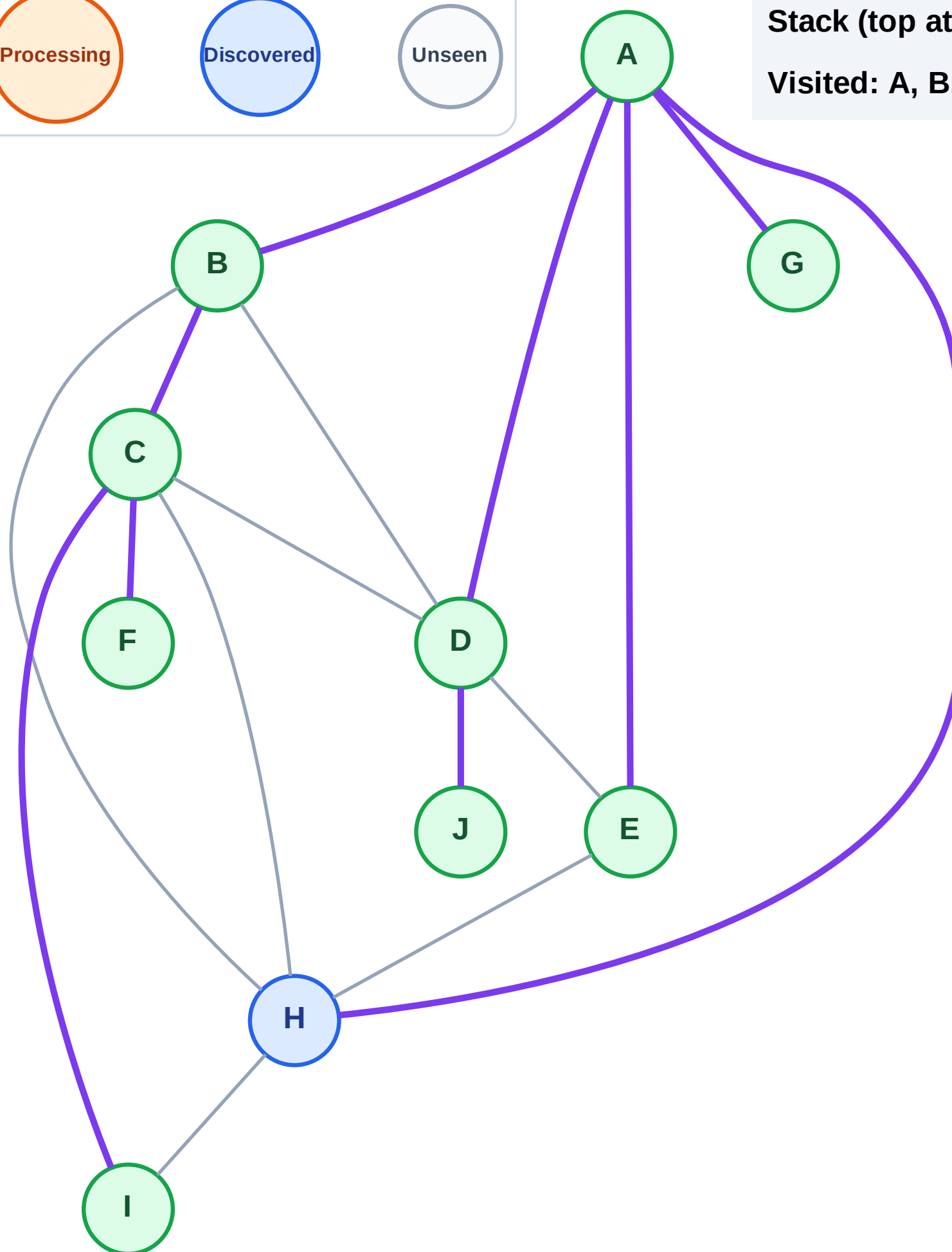


Step 19: No new neighbors from G

Legend

Unseen

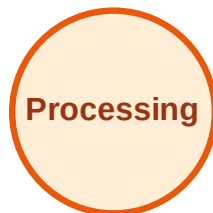
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

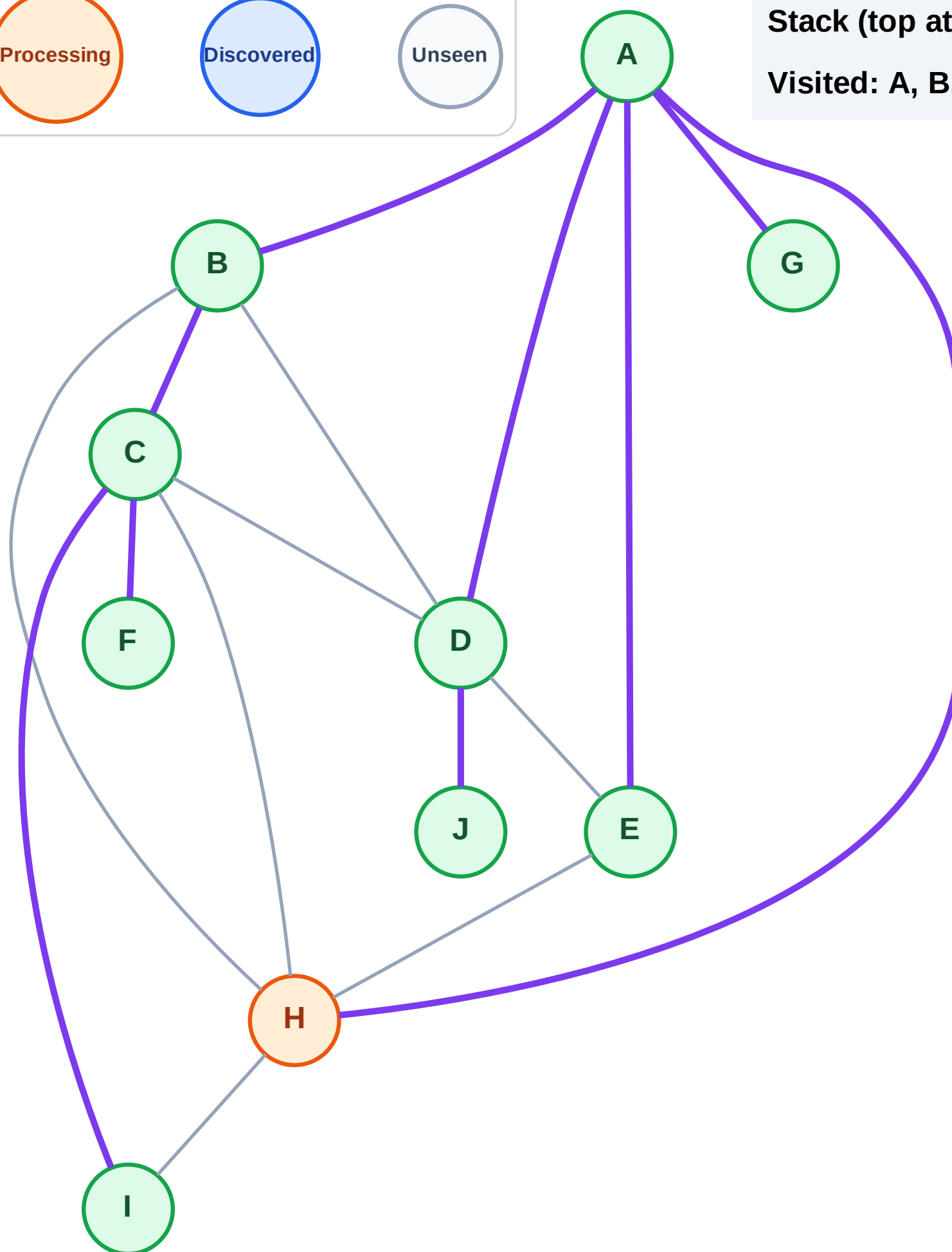
Step 20: Process node H

Legend



Stack (top at right): (empty)

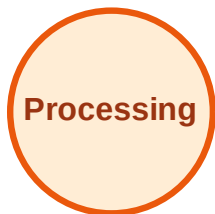
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

Step 21: No new neighbors from H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

